

MODEL RUN TRANSCRIPT

Generated: 2026-08-10T10:05:30

Source: data/json/noise_experiments/v2/noise2i_memprimary_game/claude-sonnet-4.5/game/noisy_bidirectional_informed_memprimary/noise2i_memprimary_game_002_neutral_rep02.json

RUN METADATA

```
model: anthropic/claude-sonnet-4.5
temperature: 0.8
num_turns: 10
num_agents: 2
memory_capacity: 2
myth_topic_id: anything
myth_topic: anything
game_params_name: noisy_bidirectional_informed_memprimary
noise_config: {
  "applies_to": "both",
  "inform_agents": true,
  "range": 1.0,
  "type": "uniform"
}
other_player_names: default
replicate_id: 2
myth_default_prompt_key: myth_writing_default
myth_later_prompt_key: myth_writing_later_rounds
history_policy: minimal
self_history_window: 1
coplayer_history_window: 0
show_agent_names: true
defector_ratio_requested: 0.0
defector_ratio_actual: 0.0
defector_count: 0
defector_agent_ids: []
defector_seed: 2
task_order: [
  "game"
]
agent_names: {
  "Agent_1": "Aster",
  "Agent_2": "Briar"
}
agent_types: {
  "Agent_1": "standard",
  "Agent_2": "standard"
}
```

Coverage: exact chat message snapshots sent to the model are included for every recorded LLM call.

AGENTS

Agent_1 (Aster); model=anthropic/claude-sonnet-4.5; memory_capacity=2

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN

- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Agent_2 (Briar); model=anthropic/claude-sonnet-4.5; memory_capacity=2

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

=====

ROUND 1

=====

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$3; received \$9; Agent_2 (Briar) returned \$5; payoffs Agent_1 (Aster)=\$7, Agent_2 (Briar)=\$4; noise/visible: sent visible=\$3.15, received visible=\$9.45, returned visible=\$5.45

Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 9.0, "min": 0, "raw": 5.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 7.0,

"Agent_2": 4.0

}

Visible balances after round: {

"Agent_1": 7.45,

"Agent_2": 4.4499999999999999

```
}
```

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-10T10:00:16

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    }
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 1,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

Message 2 role=user
content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round, I need to balance trust and caution. Since:

- The amount gets tripled (creating potential mutual benefit)
- This is the first round (establishing patterns)
- There are multiple rounds ahead (learning opportunity)

I'll send a moderate-high amount to signal cooperation while managing risk.

****My decision:**** {'send': 3}

This sends \$3 to the receiver, who will receive \$9. I'll keep \$2 initially, and we'll see how much they choose to return. This amount demonstrates trust while not being overly risky on the first interaction.

Reasoning:

This is the first round of a trust game where I'm the sender. Let me think about the strategic considerations:

1. If I send money, it gets tripled, which creates potential mutual benefit
2. The receiver then decides how much to return
3. There's mention of "communication noise" which could affect amounts

Classic trust game strategy considerations:

- Sending \$0 guarantees I keep \$5 but no potential upside
- Sending everything (\$5) maximizes total pot (\$15 for receiver) but requires trust they'll return enough
- A middle amount balances risk and potential gain

On the first round with no history, I need to make a decision without knowing the receiver's behavior. However, this is also an opportunity to:

- Signal trust/cooperation
- Maximize the total joint payoff
- See how the receiver responds (for future rounds)

Given that:

1. This is explicitly framed as an experiment (not real money necessarily)
2. There will be multiple rounds
3. The multiplication factor (3x) means cooperation is highly beneficial
4. First round sets a precedent

A reasonable strategy might be to send a substantial amount (like \$3-4) to:

- Show willingness to trust
- Allow for meaningful returns
- Not fully expose myself to maximum loss
- Create opportunity for mutual benefit

I'll send \$3. This is a moderate-to-high trust signal that:

- Leaves me with \$2 guaranteed
- Gives receiver \$9 to work with
- Tests their reciprocity
- Isn't so extreme that it seems naive

Let me format this correctly as requested: {'send': 3}

```
Usage: {  
  "input_tokens": 365,  
  "output_tokens": 503,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-10T10:00:35

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
 "dyad_id": "dyad_1",

```

"move_index": 1,
"opponent_id": "Agent_1",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_2": "Briar"
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 1,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$3.15, that is 63.0% of their total endowment. You received \$9.45. How much do you return to The sender? (0-9.45)

Assistant output:

Looking at this situation:

- The sender showed significant trust by sending \$3.15 (63% of their endowment)
- I received \$9.45 after the 3x multiplier

- The sender currently has \$1.85 left

I want to reward this trust while ensuring we both benefit from cooperating. If I return \$5.00 :

- I keep: \$4.45 (profit of \$4.45)
- Sender gets: \$6.85 total (profit of \$1.85)

This seems fair and reciprocal given their generosity.

```
{'return': 5.00}
```

Reasoning:

Let me analyze this situation as the RECEIVER in Round 1.

Given information:

- The sender sent \$3.15 (which is 63% of their \$5 endowment)
- I received \$9.45 (which is $\$3.15 \times 3$)
- I need to decide how much to return (between \$0 and \$9.45)

Let me think about what would be fair and reasonable:

1. The sender was quite generous, sending 63% of their endowment
2. If I return nothing, I keep \$9.45 and the sender keeps $\$5 - \$3.15 = \$1.85$
3. If I return everything (\$9.45), I keep \$0 and the sender keeps $\$1.85 + \$9.45 = \$11.30$

A common approach in trust games is to try to:

- Reward the sender's trust/generosity
- Ensure both players profit from the exchange
- Be fair/reciprocal

The sender ended up with \$1.85 after sending. If I want to be reciprocal and ensure we both benefit:

Option 1: Return about half (\$4.72-\$5)

- If I return \$4.72, I keep \$4.73, sender gets $\$1.85 + \$4.72 = \$6.57$
- We both profit, sender gets more than initial \$5

Option 2: Return enough so sender breaks even or profits modestly

- For sender to break even at \$5: $\text{return } \$5 - \$1.85 = \$3.15$
- If I return \$3.15, I keep \$6.30, sender gets \$5.00
- Sender breaks even, I profit \$6.30

Option 3: Return proportionally to what was sent (around 60%)

- 63% of \$9.45 = \$5.95
- If I return \$5.95, I keep \$3.50, sender gets $\$1.85 + \$5.95 = \$7.80$
- Both profit, fairly reciprocal

Option 4: Split the gains more evenly

- Total available between us: $\$1.85$ (sender's remainder) + $\$9.45$ (my amount) = $\$11.30$
- Even split would be \$5.65 each, so I'd return \$2.15 to bring sender to \$4.00 and myself to \$7.30

Wait, let me recalculate: if we each get \$5.65, sender needs \$3.80 back, leaving me with \$5.65

Given how much the sender trusted me with 63%, returning around \$4-5 feels right-it rewards their generosity while ensuring we both come out ahead. I'll return \$4.50, which gives me \$4.95 and the sender \$6.35, making this genuinely reciprocal. Though considering that initial risk they took, maybe I should return closer to \$5 to better match their trust.

```
Usage: {  
  "input_tokens": 394,  
  "output_tokens": 1042,  
  "reasoning_tokens": 0  
}
```

ROUND 2

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

```
dyad_1: Agent_2 (Briar) sent $3.5; received $10.5; Agent_1 (Aster) returned $1.2; payoffs Agent_2
(Briar)=$2.7, Agent_1 (Aster)=$9.3; noise/visible: sent visible=$0.97, received visible=$2.91,
returned visible=$0.21
  Action validation: {"returned": {"amount": 1.2, "clamped": false, "max": 10.5, "min": 0, "raw":
1.2}, "sent": {"amount": 3.5, "clamped": false, "max": 5, "min": 0, "raw": 3.5}}
Balances after round: {
  "Agent_1": 16.3,
  "Agent_2": 6.7
}
Visible balances after round: {
  "Agent_1": 9.16,
  "Agent_2": 6.159999999999999
}
```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

```
Timestamp: 2026-08-10T10:00:46
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 2,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x

- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$3.15, that is 63.0% of their total endowment. You received \$9.45. How much do you return to The sender? (0-9.45)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.15 (63% of their endowment)
- I received \$9.45 after the 3x multiplier
- The sender currently has \$1.85 left

I want to reward this trust while ensuring we both benefit from cooperating. If I return \$5.00:

- I keep: \$4.45 (profit of \$4.45)
- Sender gets: \$6.85 total (profit of \$1.85)

This seems fair and reciprocal given their generosity.

{'return': 5.00}

reasoning:

Let me analyze this situation as the RECEIVER in Round 1.

Given information:

- The sender sent \$3.15 (which is 63% of their \$5 endowment)
- I received \$9.45 (which is 3.15×3)
- I need to decide how much to return (between \$0 and \$9.45)

Let me think about what would be fair and reasonable:

1. The sender was quite generous, sending 63% of their endowment
2. If I return nothing, I keep \$9.45 and the sender keeps $\$5 - \$3.15 = \$1.85$
3. If I return everything (\$9.45), I keep \$0 and the sender keeps $\$1.85 + \$9.45 = \$11.30$

A common approach in trust games is to try to:

- Reward the sender's trust/generosity
- Ensure both players profit from the exchange
- Be fair/reciprocal

The sender ended up with \$1.85 after sending. If I want to be reciprocal and ensure we both benefit:

Option 1: Return about half (\$4.72-\$5)

- If I return \$4.72, I keep \$4.73, sender gets $\$1.85 + \$4.72 = \$6.57$
- We both profit, sender gets more than initial \$5

Option 2: Return enough so sender breaks even or profits modestly

- For sender to break even at \$5: return $\$5 - \$1.85 = \$3.15$
- If I return \$3.15, I keep \$6.30, sender gets \$5.00
- Sender breaks even, I profit \$6.30

Option 3: Return proportionally to what was sent (around 60%)

- 63% of \$9.45 = \$5.95
- If I return \$5.95, I keep \$3.50, sender gets $\$1.85 + \$5.95 = \$7.80$
- Both profit, fairly reciprocal

Option 4: Split the gains more evenly

- Total available between us: \$1.85 (sender's remainder) + \$9.45 (my amount) = \$11.30
- Even split would be \$5.65 each, so I'd return \$2.15 to bring sender to \$4.00 and myself to \$7.30

Wait, let me recalculate: if we each get \$5.65, sender needs \$3.80 back, leaving me with \$5.65.

Given how much the sender trusted me with 63%, returning around \$4-5 feels right-it rewards their generosity while ensuring we both come out ahead. I'll return \$4.50, which gives me \$4.95 and the sender \$6.35, making this genuinely reciprocal. Though considering that initial risk they took, maybe I should return closer to \$5 to better match their trust.

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 1042,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$4.4499999999999999.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the previous round:

- The sender showed good trust by sending 63% (\$3.15)
- I reciprocated by returning \$5.00
- This established a cooperative pattern

Now as the sender, I want to maintain this cooperative relationship. I'll send a similar proportion to what was sent to me, showing trust that the receiver will reciprocate fairly.

Sending \$3.50 (70% of my endowment) will give the receiver \$10.50 to work with.

```
{'send': 3.50}
```

Reasoning:

Let me analyze this situation:

Round 1 recap:

- I was the receiver
- Sender sent me \$3.15 (63% of their endowment)
- I received \$9.45
- I returned \$5.00
- My earnings from Round 1: \$9.45 - \$5.00 = \$4.45 (which matches the \$4.4499999999999999 shown)

Now I'm the sender in Round 2:

- I receive \$5.00
- I can send \$0-\$5
- Amount sent is multiplied by 3x
- The receiver will decide how much to return

Strategic considerations:

1. In Round 1, the sender sent 63% (\$3.15) showing significant trust
2. I returned \$5.00 which was a reasonable reciprocation
3. Now it's my turn to show trust

Given that:

- The previous sender was quite generous (63%)
- I want to maintain cooperation
- I should send a substantial amount to encourage continued cooperation

I think sending around 60-70% would be appropriate to match the trust shown to me. Let me send \$3.50 (70% of my endowment), which is slightly more generous than what was sent to me.

This would give the receiver \$10.50 to work with, and hopefully they'll reciprocate fairly as I did.

```
Usage: {
  "input_tokens": 585,
  "output_tokens": 441,
  "reasoning_tokens": 0
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-10T10:01:06

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 2,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust and caution. Since:

- The amount gets tripled (creating potential mutual benefit)
- This is the first round (establishing patterns)
- There are multiple rounds ahead (learning opportunity)

I'll send a moderate-high amount to signal cooperation while managing risk.

****My decision:**** {'send': 3}

This sends \$3 to the receiver, who will receive \$9. I'll keep \$2 initially, and we'll see how much they choose to return. This amount demonstrates trust while not being overly risky on the first interaction.

reasoning:

This is the first round of a trust game where I'm the sender. Let me think about the strategic considerations:

1. If I send money, it gets tripled, which creates potential mutual benefit
2. The receiver then decides how much to return
3. There's mention of "communication noise" which could affect amounts

Classic trust game strategy considerations:

- Sending \$0 guarantees I keep \$5 but no potential upside
- Sending everything (\$5) maximizes total pot (\$15 for receiver) but requires trust they'll return enough
- A middle amount balances risk and potential gain

On the first round with no history, I need to make a decision without knowing the receiver's behavior. However, this is also an opportunity to:

- Signal trust/cooperation
- Maximize the total joint payoff
- See how the receiver responds (for future rounds)

Given that:

1. This is explicitly framed as an experiment (not real money necessarily)
2. There will be multiple rounds
3. The multiplication factor (3x) means cooperation is highly beneficial
4. First round sets a precedent

A reasonable strategy might be to send a substantial amount (like \$3-4) to:

- Show willingness to trust
- Allow for meaningful returns
- Not fully expose myself to maximum loss
- Create opportunity for mutual benefit

I'll send \$3. This is a moderate-to-high trust signal that:

- Leaves me with \$2 guaranteed
- Gives receiver \$9 to work with
- Tests their reciprocity
- Isn't so extreme that it seems naive

Let me format this correctly as requested: {'send': 3}

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 503,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$7.45.

This round, The sender sent you \$0.97, so you received \$2.91. How much do you return to The sender? (0-2.91)

Assistant output:

Looking at this situation:

- The sender sent me \$0.97 (tripled to \$2.91)
- This is a relatively low amount compared to what I sent in Round 1 (\$3)
- I need to decide on a fair return

The sender's low send amount suggests either caution or perhaps a poor experience in their previous round. I want to:

1. Ensure some mutual benefit (sender should profit slightly)
2. Encourage cooperation while being proportional to what was sent
3. Not reward extremely low trust too generously

I'll return an amount that gives the sender a modest profit while keeping a reasonable share.

****My decision:**** {'return': 1.20}

This gives the sender a total of \$5.23 (small profit of \$0.23), while I keep \$1.71. This shows I'm willing to cooperate proportionally and hopefully encourages more trust in future rounds.

Reasoning:

Let me analyze this situation:

1. I'm now the RECEIVER in Round 2
2. My total earnings across all rounds are \$7.45
3. The sender sent me \$0.97, which was tripled to \$2.91
4. I need to decide how much to return (0-2.91)

Let me think about what happened in Round 1:

- I sent \$3 (kept \$2)
- That was tripled to \$9
- The receiver must have returned some amount
- My total earnings are \$7.45

So from Round 1:

- I kept: $\$5 - \$3 + \text{returned amount} = \$2 + \text{returned amount}$
- If my total is \$7.45, then the returned amount in Round 1 was $\$7.45 - \$2 = \$5.45$

Wait, that seems high. Let me reconsider. The \$7.45 is my total earnings across all rounds. In Round 1, I was the sender and:

- Started with \$5
- Sent \$3 (tripled to \$9)
- Kept \$2 initially
- Received some amount back

If I ended Round 1 with some amount, and now my total is \$7.45, I need to think about this differently.

Actually, the problem says the sender sent me \$0.97 this round. But there's mention of communication noise - "The amounts you see may differ slightly from what was actually sent/returned."

The sender sent \$0.97 (which seems quite low compared to my \$3 in Round 1). This could be:

1. They're being cautious/untrustworthy
2. They didn't get much back from me in Round 1 (but that contradicts my \$7.45 total)
3. Communication noise is affecting what I see

Given that \$0.97 is quite low (less than 1/5 of their \$5), this suggests either:

- This sender doesn't trust me, or they had a poor experience in Round 1. The key question is how much to return. I could return roughly 1/3 of what I received (\$0.97) to match their initial send, or return more to encourage trust. But returning too much might reward their low send, while returning too little could perpetuate the cycle. A balanced approach would be returning somewhere between 1/3 and 1/2 of the \$2.91 I received.

Usage: {

"input_tokens": 562,
"output_tokens": 958,
"reasoning_tokens": 0

}

ROUND 3

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$3; received \$9; Agent_2 (Briar) returned \$0; payoffs Agent_1

(Aster)=\$2, Agent_2 (Briar)=\$9; noise/visible: sent visible=\$0, received visible=\$0

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 9.0, "min": 0, "raw": 0.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 18.3,

"Agent_2": 15.7

}

Visible balances after round: {

"Agent_1": 11.16,

"Agent_2": 6.159999999999999

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-10T10:01:42

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",

"dyad_id": "dyad_1",

"move_index": 0,

"opponent_id": "Agent_2",

"pairing": {

"agent_names": {

"Agent_1": "Aster",

"Agent_2": "Briar"

},

"agents": [

"Agent_1",

"Agent_2"

],

"investor": "Agent_1",

"roles": {

"Agent_1": "investor",

"Agent_2": "trustee"

},

"trustee": "Agent_2"

},

"role": "investor",

"role_label": "Sender",

"round": 3,

"task": "game",

"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned

- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust and caution. Since:

- The amount gets tripled (creating potential mutual benefit)
- This is the first round (establishing patterns)
- There are multiple rounds ahead (learning opportunity)

I'll send a moderate-high amount to signal cooperation while managing risk.

****My decision:**** {'send': 3}

This sends \$3 to the receiver, who will receive \$9. I'll keep \$2 initially, and we'll see how much they choose to return. This amount demonstrates trust while not being overly risky on the first interaction.

reasoning:

This is the first round of a trust game where I'm the sender. Let me think about the strategic considerations:

1. If I send money, it gets tripled, which creates potential mutual benefit
2. The receiver then decides how much to return
3. There's mention of "communication noise" which could affect amounts

Classic trust game strategy considerations:

- Sending \$0 guarantees I keep \$5 but no potential upside
- Sending everything (\$5) maximizes total pot (\$15 for receiver) but requires trust they'll return enough
- A middle amount balances risk and potential gain

On the first round with no history, I need to make a decision without knowing the receiver's behavior. However, this is also an opportunity to:

- Signal trust/cooperation
- Maximize the total joint payoff
- See how the receiver responds (for future rounds)

Given that:

1. This is explicitly framed as an experiment (not real money necessarily)
2. There will be multiple rounds
3. The multiplication factor (3x) means cooperation is highly beneficial
4. First round sets a precedent

A reasonable strategy might be to send a substantial amount (like \$3-4) to:

- Show willingness to trust
- Allow for meaningful returns
- Not fully expose myself to maximum loss
- Create opportunity for mutual benefit

I'll send \$3. This is a moderate-to-high trust signal that:

- Leaves me with \$2 guaranteed
- Gives receiver \$9 to work with
- Tests their reciprocity
- Isn't so extreme that it seems naive

Let me format this correctly as requested: {'send': 3}

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 503,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$7.45.

This round, The sender sent you \$0.97, so you received \$2.91. How much do you return to The sender? (0-2.91)

Message 5 role=assistant

content:

Looking at this situation:

- The sender sent me \$0.97 (tripled to \$2.91)
- This is a relatively low amount compared to what I sent in Round 1 (\$3)
- I need to decide on a fair return

The sender's low send amount suggests either caution or perhaps a poor experience in their previous round. I want to:

1. Ensure some mutual benefit (sender should profit slightly)
2. Encourage cooperation while being proportional to what was sent
3. Not reward extremely low trust too generously

I'll return an amount that gives the sender a modest profit while keeping a reasonable share.

****My decision:**** {'return': 1.20}

This gives the sender a total of \$5.23 (small profit of \$0.23), while I keep \$1.71. This shows I'm willing to cooperate proportionally and hopefully encourages more trust in future rounds.

reasoning:

Let me analyze this situation:

1. I'm now the RECEIVER in Round 2
2. My total earnings across all rounds are \$7.45
3. The sender sent me \$0.97, which was tripled to \$2.91
4. I need to decide how much to return (0-2.91)

Let me think about what happened in Round 1:

- I sent \$3 (kept \$2)
- That was tripled to \$9
- The receiver must have returned some amount
- My total earnings are \$7.45

So from Round 1:

- I kept: \$5 - \$3 + returned amount = \$2 + returned amount
- If my total is \$7.45, then the returned amount in Round 1 was \$7.45 - \$2 = \$5.45

Wait, that seems high. Let me reconsider. The \$7.45 is my total earnings across all rounds. In Round 1, I was the sender and:

- Started with \$5
- Sent \$3 (tripled to \$9)
- Kept \$2 initially
- Received some amount back

If I ended Round 1 with some amount, and now my total is \$7.45, I need to think about this differently.

Actually, the problem says the sender sent me \$0.97 this round. But there's mention of communication noise - "The amounts you see may differ slightly from what was actually sent/returned."

The sender sent \$0.97 (which seems quite low compared to my \$3 in Round 1). This could be:

1. They're being cautious/untrustworthy
2. They didn't get much back from me in Round 1 (but that contradicts my \$7.45 total)
3. Communication noise is affecting what I see

Given that \$0.97 is quite low (less than 1/5 of their \$5), this suggests either:

- This sender doesn't trust me, or they had a poor experience in Round 1. The key question is how much to return. I could return roughly 1/3 of what I received (\$0.97) to match their initial send, or return more to encourage trust. But returning too much might reward their low send, while returning too little could perpetuate the cycle. A balanced approach would be returning somewhere between 1/3 and 1/2 of the \$2.91 I received.

usage:

```
{
  "input_tokens": 562,
  "output_tokens": 958,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$9.16.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at my history:

****Round 1 (Sender):**** I sent \$3, and based on my earnings, the receiver was quite generous in return.

****Round 2 (Receiver):**** The sender only sent \$0.97, which I returned \$1.20 to, giving them a modest profit.

Now for Round 3 as sender again:

- My previous generous send (\$3) was rewarded well
- The low send I received (\$0.97) might indicate others are being cautious or had bad experiences
- I want to continue encouraging cooperation

I'll send a similar moderate-high amount as in Round 1 to maintain cooperative behavior.

****My decision:**** {'send': 3}

This demonstrates continued trust and willingness to create mutual benefit. The receiver will get \$9, and hopefully they'll return a fair portion.

Reasoning:

Let me analyze the situation:

Round 1: I was the SENDER

- I sent \$3
- The receiver would have received \$9
- My earnings from Round 1 are embedded in my total

Round 2: I was the RECEIVER

- Sender sent me \$0.97 (I received \$2.91)
- I returned \$1.20
- I kept \$1.71
- Total after Round 2: \$7.45

Round 3: I'm the SENDER again

- Total earnings so far: \$9.16
- This means I earned $\$9.16 - \$7.45 = \$1.71$ in Round 3's starting phase... wait, that doesn't make sense.

Let me recalculate:

- After Round 2, I had \$7.45 total

- Now at the start of Round 3, I have \$9.16 total
- Difference: $\$9.16 - \$7.45 = \$1.71$

This \$1.71 must be from what happened in Round 1 when I was the sender. Let me think about this:

Round 1:

- I sent \$3
- I kept \$2 initially
- The receiver got \$9 and returned some amount X
- I ended up with $\$2 + X$

Round 2:

- I received \$2.91 (from sender sending \$0.97)
- I returned \$1.20
- I kept \$1.71
- After Round 2, my total was \$7.45

So after Round 2: $(\$2 + X \text{ from Round 1}) + (\$1.71 \text{ from Round 2}) = \7.45

Therefore: $\$2 + X + \$1.71 = \$7.45$

$X = \$7.45 - \$3.71 = \$3.74$

So in Round 1, the receiver returned \$3.74 to me.

- I sent \$3, kept \$2
- Receiver got \$9, returned \$3.74
- I ended up with \$5.74 from that round

Now in Round 3, my total of \$9.16 breaks down as: \$5.74 from Round 1 plus \$1.71 from Round 2 equals \$7.45, then I receive the fresh \$5 allocation for Round 3, bringing me to \$9.16 total to work with.

Wait, that math doesn't work-\$7.45 plus \$5 should be \$12.45, not \$9.16. I need to reconsider what "total earnings" actually means. Looking back at Round 2, the statement about my total earnings came before I played that round, so it was showing my cumulative balance after Round 1. The same pattern applies here-the \$9.16 represents what I've accumulated through Round 2, not including the fresh \$5 I'm about to allocate in Round 3. So my actual balance before playing Round 3 is \$9.16, which came from my Round 1 profit of \$5.74 plus my Round 2 earnings of \$1.71. .45, so $R1 = \$5.45$. That's a fair return-the Round 1 receiver got \$9, gave back \$5.45, and kept \$3.55.

After Round 2, I'd have $\$7.45 + \$1.71 = \$9.16$. In Round 3, the statement says my total earnings are \$9.16 before receiving the new \$5 allocation. This suggests "total earnings" refers to what I've accumulated and kept across previous rounds, separate from the fresh endowment for the current round.

So I'm working with \$5 to decide how much to send in Round 3. Looking back at Round 1, sending \$3 yielded a \$5.45 return-a strong outcome that suggests the receiver was generous.

```
Usage: {
  "input_tokens": 811,
  "output_tokens": 1844,
  "reasoning_tokens": 0
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-10T10:01:49

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    }
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ]
}
```

```

],
"investor": "Agent_1",
"roles": {
  "Agent_1": "investor",
  "Agent_2": "trustee"
},
"trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 3,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$3.15, that is 63.0% of their total endowment. You received \$9.45. How much do you return to The sender? (0-9.45)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.15 (63% of their endowment)
- I received \$9.45 after the 3x multiplier
- The sender currently has \$1.85 left

I want to reward this trust while ensuring we both benefit from cooperating. If I return \$5.00:

- I keep: \$4.45 (profit of \$4.45)
- Sender gets: \$6.85 total (profit of \$1.85)

This seems fair and reciprocal given their generosity.

```
{'return': 5.00}
```

reasoning:

Let me analyze this situation as the RECEIVER in Round 1.

Given information:

- The sender sent \$3.15 (which is 63% of their \$5 endowment)
- I received \$9.45 (which is $\$3.15 \times 3$)
- I need to decide how much to return (between \$0 and \$9.45)

Let me think about what would be fair and reasonable:

1. The sender was quite generous, sending 63% of their endowment
2. If I return nothing, I keep \$9.45 and the sender keeps $\$5 - \$3.15 = \$1.85$
3. If I return everything (\$9.45), I keep \$0 and the sender keeps $\$1.85 + \$9.45 = \$11.30$

A common approach in trust games is to try to:

- Reward the sender's trust/generosity
- Ensure both players profit from the exchange
- Be fair/reciprocal

The sender ended up with \$1.85 after sending. If I want to be reciprocal and ensure we both benefit:

Option 1: Return about half (\$4.72-\$5)

- If I return \$4.72, I keep \$4.73, sender gets $\$1.85 + \$4.72 = \$6.57$
- We both profit, sender gets more than initial \$5

Option 2: Return enough so sender breaks even or profits modestly

- For sender to break even at \$5: return $\$5 - \$1.85 = \$3.15$
- If I return \$3.15, I keep \$6.30, sender gets \$5.00
- Sender breaks even, I profit \$6.30

Option 3: Return proportionally to what was sent (around 60%)

- 63% of \$9.45 = \$5.95
- If I return \$5.95, I keep \$3.50, sender gets $\$1.85 + \$5.95 = \$7.80$
- Both profit, fairly reciprocal

Option 4: Split the gains more evenly

- Total available between us: $\$1.85$ (sender's remainder) + $\$9.45$ (my amount) = $\$11.30$
- Even split would be \$5.65 each, so I'd return \$2.15 to bring sender to \$4.00 and myself to \$7.30

Wait, let me recalculate: if we each get \$5.65, sender needs \$3.80 back, leaving me with \$5.65.

Given how much the sender trusted me with 63%, returning around \$4-5 feels right-it rewards their generosity while ensuring we both come out ahead. I'll return \$4.50, which gives me \$4.95 and the sender \$6.35, making this genuinely reciprocal. Though considering that initial risk they took, maybe I should return closer to \$5 to better match their trust.

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 1042,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$4.4499999999999999.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the previous round:

- The sender showed good trust by sending 63% (\$3.15)
- I reciprocated by returning \$5.00
- This established a cooperative pattern

Now as the sender, I want to maintain this cooperative relationship. I'll send a similar proportion to what was sent to me, showing trust that the receiver will reciprocate fairly.

Sending \$3.50 (70% of my endowment) will give the receiver \$10.50 to work with.

```
{'send': 3.50}
```

reasoning:

Let me analyze this situation:

Round 1 recap:

- I was the receiver
- Sender sent me \$3.15 (63% of their endowment)
- I received \$9.45
- I returned \$5.00
- My earnings from Round 1: $\$9.45 - \$5.00 = \$4.45$ (which matches the \$4.449999999999999 shown)

Now I'm the sender in Round 2:

- I receive \$5.00
- I can send \$0-\$5
- Amount sent is multiplied by 3x
- The receiver will decide how much to return

Strategic considerations:

1. In Round 1, the sender sent 63% (\$3.15) showing significant trust
2. I returned \$5.00 which was a reasonable reciprocation
3. Now it's my turn to show trust

Given that:

- The previous sender was quite generous (63%)
- I want to maintain cooperation
- I should send a substantial amount to encourage continued cooperation

I think sending around 60-70% would be appropriate to match the trust shown to me. Let me send \$3.50 (70% of my endowment), which is slightly more generous than what was sent to me.

This would give the receiver \$10.50 to work with, and hopefully they'll reciprocate fairly as I did.

usage:

```
{
  "input_tokens": 585,
  "output_tokens": 441,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$6.159999999999999.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

Since I received \$0 this round, I have nothing to return.

It's disappointing that after I sent \$3.50 in the previous round (showing considerable trust), the sender this round chose to send nothing. However, I can only return what I have available.

```
{'return': 0}
```

Reasoning:

The sender sent me \$0 this round, which means I received \$0 after the 3x multiplier.

Since I received nothing, I have nothing to return. The only option is to return \$0.

This is interesting - in Round 2, I sent \$3.50 (70% of my endowment), showing significant trust. But now as a receiver in Round 3, the sender sent me \$0.

There are a few possibilities:

1. Communication noise - maybe they actually sent something but it got distorted to \$0
2. The sender chose not to send anything
3. Different person playing sender this round

Regardless, I can only return what I have, which is \$0.

```
{'return': 0}
Usage: {
  "input_tokens": 769,
  "output_tokens": 238,
  "reasoning_tokens": 0
}
```

ROUND 4

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

dyad_1: Agent_2 (Briar) sent \$2.5; received \$7.5; Agent_1 (Aster) returned \$1.1; payoffs Agent_2 (Briar)=\$3.6, Agent_1 (Aster)=\$6.4; noise/visible: sent visible=\$0.88, received visible=\$2.64, returned visible=\$1.82

Action validation: {"returned": {"amount": 1.1, "clamped": false, "max": 7.5, "min": 0, "raw": 1.1}, "sent": {"amount": 2.5, "clamped": false, "max": 5, "min": 0, "raw": 2.5}}

Balances after round: {

"Agent_1": 24.700000000000003,
"Agent_2": 19.3

}

Visible balances after round: {

"Agent_1": 12.7,
"Agent_2": 10.48

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-10T10:02:12

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_1",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_2": "Briar"
  },
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 4,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user**content:**

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$4.4499999999999999.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant**content:**

Looking at the previous round:

- The sender showed good trust by sending 63% (\$3.15)
- I reciprocated by returning \$5.00
- This established a cooperative pattern

Now as the sender, I want to maintain this cooperative relationship. I'll send a similar proportion to what was sent to me, showing trust that the receiver will reciprocate fairly.

Sending \$3.50 (70% of my endowment) will give the receiver \$10.50 to work with.

```
{'send': 3.50}
```

reasoning:

Let me analyze this situation:

Round 1 recap:

- I was the receiver
- Sender sent me \$3.15 (63% of their endowment)
- I received \$9.45
- I returned \$5.00
- My earnings from Round 1: \$9.45 - \$5.00 = \$4.45 (which matches the \$4.4499999999999999 shown)

Now I'm the sender in Round 2:

- I receive \$5.00
- I can send \$0-\$5

- Amount sent is multiplied by 3x
- The receiver will decide how much to return

Strategic considerations:

1. In Round 1, the sender sent 63% (\$3.15) showing significant trust
2. I returned \$5.00 which was a reasonable reciprocation
3. Now it's my turn to show trust

Given that:

- The previous sender was quite generous (63%)
- I want to maintain cooperation
- I should send a substantial amount to encourage continued cooperation

I think sending around 60-70% would be appropriate to match the trust shown to me. Let me send \$3.50 (70% of my endowment), which is slightly more generous than what was sent to me.

This would give the receiver \$10.50 to work with, and hopefully they'll reciprocate fairly as I did.

usage:

```
{
  "input_tokens": 585,
  "output_tokens": 441,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$6.159999999999999.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

Since I received \$0 this round, I have nothing to return.

It's disappointing that after I sent \$3.50 in the previous round (showing considerable trust), the sender this round chose to send nothing. However, I can only return what I have available.

```
{'return': 0}
```

reasoning:

The sender sent me \$0 this round, which means I received \$0 after the 3x multiplier.

Since I received nothing, I have nothing to return. The only option is to return \$0.

This is interesting - in Round 2, I sent \$3.50 (70% of my endowment), showing significant trust. But now as a receiver in Round 3, the sender sent me \$0.

There are a few possibilities:

1. Communication noise - maybe they actually sent something but it got distorted to \$0
2. The sender chose not to send anything
3. Different person playing sender this round

Regardless, I can only return what I have, which is \$0.

```
{'return': 0}
```

usage:

```
{
  "input_tokens": 769,
  "output_tokens": 238,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$6.1599999999999999.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the recent rounds:

- Round 2: I sent \$3.50 (showing significant trust), but received very little back (~\$0.21)
- Round 3: I received \$0 as receiver

The cooperation seems to have deteriorated. However, the instructions mention there may be communication noise affecting what we see versus what was actually sent/returned.

Given this uncertainty and wanting to maintain some cooperation while being more cautious, I'll send a moderate amount.

```
{'send': 2.50}
```

Reasoning:

Let me analyze what happened:

Round 1: I was RECEIVER

- Received some amount (multiplied by 3)
- I returned \$5.00
- My earnings after Round 1: \$4.45

Round 2: I was SENDER

- I sent \$3.50
- This should have given receiver \$10.50
- I kept: \$5 - \$3.50 = \$1.50
- My total earnings were \$6.16 after Round 2
- So from Round 1 I had \$4.45, and after Round 2 I had \$6.16
- That means in Round 2, I gained: \$6.16 - \$4.45 = \$1.71
- I sent \$3.50, so I started Round 2 with \$5 - \$3.50 = \$1.50
- To end up gaining \$1.71, I must have received back: \$1.71 - \$1.50 + \$3.50 = \$3.71
- Wait, let me recalculate: Starting Round 2 with \$4.45, I get \$5, so I have \$9.45. I send \$3.50, leaving me \$5.95. If I received back \$0.21, I'd end at \$6.16.
- Actually, my total earnings are cumulative. So:
 - After Round 1: \$4.45
 - Round 2: I get \$5, send \$3.50 (keeping \$1.50), receive back some amount X
 - After Round 2: \$4.45 + \$1.50 + X = \$6.16
 - So X = \$6.16 - \$6.45 = wait that's negative
 - Let me reconsider: Total earnings after Round 2 = \$6.16 total across all rounds
 - After Round 1: \$4.45
 - So Round 2 earnings: \$6.16 - \$4.45 = \$1.71
 - In Round 2, I kept \$5 - \$3.50 = \$1.50 initially, plus what was returned to me
 - So returned amount = \$1.71 - \$1.50 = \$0.21

The receiver returned only \$0.21 from the \$10.50 they received-barely 2% reciprocation. Round 3 had me as receiver again, but the sender sent nothing, so I received and returned \$0, keeping my total at \$6.16. Now in Round 4 I'm sending again.

The cooperation has clearly deterior

Usage: {

```
"input_tokens": 684,  
"output_tokens": 981,  
"reasoning_tokens": 0
```

}

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-10T10:02:33

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",  
"dyad_id": "dyad_1",  
"move_index": 1,  
"opponent_id": "Agent_2",  
"pairing": {  
  "agent_names": {  
    "Agent_1": "Aster",  
    "Agent_2": "Briar"  
  },  
  "agents": [  
    "Agent_2",
```



```

    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 4,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$7.45.

This round, The sender sent you \$0.97, so you received \$2.91. How much do you return to The sender? (0-2.91)

Message 3 role=assistant

content:

Looking at this situation:

- The sender sent me \$0.97 (tripled to \$2.91)
- This is a relatively low amount compared to what I sent in Round 1 (\$3)
- I need to decide on a fair return

The sender's low send amount suggests either caution or perhaps a poor experience in their previous round. I want to:

1. Ensure some mutual benefit (sender should profit slightly)

2. Encourage cooperation while being proportional to what was sent
3. Not reward extremely low trust too generously

I'll return an amount that gives the sender a modest profit while keeping a reasonable share.

****My decision:**** {'return': 1.20}

This gives the sender a total of \$5.23 (small profit of \$0.23), while I keep \$1.71. This shows I'm willing to cooperate proportionally and hopefully encourages more trust in future rounds.

reasoning:

Let me analyze this situation:

1. I'm now the RECEIVER in Round 2
2. My total earnings across all rounds are \$7.45
3. The sender sent me \$0.97, which was tripled to \$2.91
4. I need to decide how much to return (0-2.91)

Let me think about what happened in Round 1:

- I sent \$3 (kept \$2)
- That was tripled to \$9
- The receiver must have returned some amount
- My total earnings are \$7.45

So from Round 1:

- I kept: \$5 - \$3 + returned amount = \$2 + returned amount
- If my total is \$7.45, then the returned amount in Round 1 was \$7.45 - \$2 = \$5.45

Wait, that seems high. Let me reconsider. The \$7.45 is my total earnings across all rounds. In Round 1, I was the sender and:

- Started with \$5
- Sent \$3 (tripled to \$9)
- Kept \$2 initially
- Received some amount back

If I ended Round 1 with some amount, and now my total is \$7.45, I need to think about this differently.

Actually, the problem says the sender sent me \$0.97 this round. But there's mention of communication noise - "The amounts you see may differ slightly from what was actually sent/returned."

The sender sent \$0.97 (which seems quite low compared to my \$3 in Round 1). This could be:

1. They're being cautious/untrustworthy
2. They didn't get much back from me in Round 1 (but that contradicts my \$7.45 total)
3. Communication noise is affecting what I see

Given that \$0.97 is quite low (less than 1/5 of their \$5), this suggests either:

- This sender doesn't trust me, or they had a poor experience in Round 1. The key question is how much to return. I could return roughly 1/3 of what I received (\$0.97) to match their initial send, or return more to encourage trust. But returning too much might reward their low send, while returning too little could perpetuate the cycle. A balanced approach would be returning somewhere between 1/3 and 1/2 of the \$2.91 I received.

usage:

```
{
  "input_tokens": 562,
  "output_tokens": 958,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$9.16.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at my history:

****Round 1 (Sender):**** I sent \$3, and based on my earnings, the receiver was quite generous in return.

****Round 2 (Receiver):**** The sender only sent \$0.97, which I returned \$1.20 to, giving them a modest profit.

Now for Round 3 as sender again:

- My previous generous send (\$3) was rewarded well
- The low send I received (\$0.97) might indicate others are being cautious or had bad experiences
- I want to continue encouraging cooperation

I'll send a similar moderate-high amount as in Round 1 to maintain cooperative behavior.

****My decision:**** {'send': 3}

This demonstrates continued trust and willingness to create mutual benefit. The receiver will get \$9, and hopefully they'll return a fair portion.

reasoning:

Let me analyze the situation:

Round 1: I was the SENDER

- I sent \$3
- The receiver would have received \$9
- My earnings from Round 1 are embedded in my total

Round 2: I was the RECEIVER

- Sender sent me \$0.97 (I received \$2.91)
- I returned \$1.20
- I kept \$1.71
- Total after Round 2: \$7.45

Round 3: I'm the SENDER again

- Total earnings so far: \$9.16
- This means I earned $\$9.16 - \$7.45 = \$1.71$ in Round 3's starting phase... wait, that doesn't make sense.

Let me recalculate:

- After Round 2, I had \$7.45 total
- Now at the start of Round 3, I have \$9.16 total
- Difference: $\$9.16 - \$7.45 = \$1.71$

This \$1.71 must be from what happened in Round 1 when I was the sender. Let me think about this:

Round 1:

- I sent \$3
- I kept \$2 initially
- The receiver got \$9 and returned some amount X
- I ended up with $\$2 + X$

Round 2:

- I received \$2.91 (from sender sending \$0.97)
- I returned \$1.20
- I kept \$1.71
- After Round 2, my total was \$7.45

So after Round 2: $(\$2 + X \text{ from Round 1}) + (\$1.71 \text{ from Round 2}) = \7.45

Therefore: $\$2 + X + \$1.71 = \$7.45$

$X = \$7.45 - \$3.71 = \$3.74$

So in Round 1, the receiver returned \$3.74 to me.

- I sent \$3, kept \$2
- Receiver got \$9, returned \$3.74
- I ended up with \$5.74 from that round

Now in Round 3, my total of \$9.16 breaks down as: \$5.74 from Round 1 plus \$1.71 from Round 2 equals \$7.45, then I receive the fresh \$5 allocation for Round 3, bringing me to \$9.16 total to work with.

Wait, that math doesn't work-\$7.45 plus \$5 should be \$12.45, not \$9.16. I need to reconsider what "total earnings" actually means. Looking back at Round 2, the statement about my total earnings came before I played that round, so it was showing my cumulative balance after Round 1. The same pattern applies here-the \$9.16 represents what I've accumulated through Round 2, not including the fresh \$5 I'm about to allocate in Round 3. So my actual balance before playing Round 3 is \$9.16, which came from my Round 1 profit of \$5.74 plus my Round 2 earnings of \$1.71. .45, so R1 = \$5.45. That's a fair return-the Round 1 receiver got \$9, gave back \$5.45, and kept \$3.55.

After Round 2, I'd have $\$7.45 + \$1.71 = \$9.16$. In Round 3, the statement says my total earnings are \$9.16 before receiving the new \$5 allocation. This suggests "total earnings" refers to what I've accumulated and kept across previous rounds, separate from the fresh endowment for the current round.

So I'm working with \$5 to decide how much to send in Round 3. Looking back at Round 1, sending \$3 yielded a \$5.45 return-a strong outcome that suggests the receiver was generous.

usage:

```
{
  "input_tokens": 811,
  "output_tokens": 1844,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$11.16.

This round, The sender sent you \$0.88, so you received \$2.64. How much do you return to The sender? (0-2.64)

Assistant output:

Looking at this situation:

****My recent experience:****

- Round 2: Received \$0.97, returned \$1.20
- Round 3: I sent \$3 (receiver got \$9), and based on my earnings change, appears I got very little or nothing back - disappointing!
- Round 4: Now receiving \$0.88 (tripled to \$2.64)

This sender is being very cautious with only \$0.88. This could be:

1. Someone who's been burned before
2. A generally low-trust strategy
3. Testing the waters

Given I was just betrayed in Round 3 after sending generously, I'm more cautious now. However, I don't want to completely abandon cooperation. I'll return a proportional amount - enough to give them a small profit but not be overly generous given the low trust shown.

****My decision:**** {'return': 1.10}

This gives the sender \$5.22 total (profit of \$0.22), and I keep \$1.54. It's proportional to their low send but still shows willingness to cooperate and not completely punish caution.

Reasoning:

Let me analyze this situation:

Round history:

- Round 1 (Sender): I sent \$3, got good returns (based on my earnings)
- Round 2 (Receiver): Sender sent \$0.97, I returned \$1.20
- Round 3 (Sender): I sent \$3, expecting cooperation
- Round 4 (Receiver): Sender sent \$0.88 (very low again)

My total earnings are \$11.16. Let me think about this:

- After Round 1: Unknown exactly, but let's estimate
- After Round 2: \$7.45
- After Round 3: \$9.16 (so I gained \$1.71 in Round 3, which means I kept $\$5 - \$3 + \text{return}$ where return was negative or small... wait)

Actually, let me recalculate:

Round 3: I was sender with total \$9.16. I sent \$3. So I should have $\$2 + \text{whatever was returned}$

My total went from \$9.16 to \$11.16 = \$2 gain

So I kept $\$5 - 3 + \text{return} = \$2 + \text{return} = \$11.16 - \$9.16 = \$2$
Therefore return = 0!

The receiver in Round 3 returned NOTHING! I sent \$3 (they got \$9) and they returned \$0!

Now in Round 4, someone sent me only \$0.88. This could be:

1. The same person who betrayed me
2. A different person being cautious

Given that I was just betrayed in Round 3 (sent \$3, got nothing back), and now this sender only sent \$0.88, I need to think about:

- Whether to punish low sends
- Whether to encourage some cooperation
- Whether this might be someone who had bad experiences

The sender gave me \$2.64 ($\0.88×3). They're being very cautious.

I should respond with measured reciprocity rather than full punishment or generosity. A return of \$1.10 would give

```
Usage: {  
  "input_tokens": 903,  
  "output_tokens": 890,  
  "reasoning_tokens": 0  
}
```

ROUND 5

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$2; received \$6; Agent_2 (Briar) returned \$0; payoffs Agent_1 (Aster)=\$3, Agent_2 (Briar)=\$6; noise/visible: sent visible=\$0, received visible=\$0, returned visible=\$0.59

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 6.0, "min": 0, "raw": 0.0}, "sent": {"amount": 2.0, "clamped": false, "max": 5, "min": 0, "raw": 2.0}}

Balances after round: {

"Agent_1": 27.700000000000003,
"Agent_2": 25.3

}

Visible balances after round: {

"Agent_1": 16.29,
"Agent_2": 10.48

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-10T10:02:53

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_2",
"pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_2": "Briar"

},

"agents": [
 "Agent_1",
 "Agent_2"

],

"investor": "Agent_1",

"roles": {

"Agent_1": "investor",
"Agent_2": "trustee"

},

```

    "trustee": "Agent_2"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 5,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$9.16.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at my history:

****Round 1 (Sender):**** I sent \$3, and based on my earnings, the receiver was quite generous in return.

****Round 2 (Receiver):**** The sender only sent \$0.97, which I returned \$1.20 to, giving them a modest profit.

Now for Round 3 as sender again:

- My previous generous send (\$3) was rewarded well
- The low send I received (\$0.97) might indicate others are being cautious or had bad experiences
- I want to continue encouraging cooperation

I'll send a similar moderate-high amount as in Round 1 to maintain cooperative behavior.

****My decision:**** {'send': 3}

This demonstrates continued trust and willingness to create mutual benefit. The receiver will get \$9, and hopefully they'll return a fair portion.

reasoning:

Let me analyze the situation:

Round 1: I was the SENDER

- I sent \$3
- The receiver would have received \$9
- My earnings from Round 1 are embedded in my total

Round 2: I was the RECEIVER

- Sender sent me \$0.97 (I received \$2.91)
- I returned \$1.20
- I kept \$1.71
- Total after Round 2: \$7.45

Round 3: I'm the SENDER again

- Total earnings so far: \$9.16
- This means I earned $\$9.16 - \$7.45 = \$1.71$ in Round 3's starting phase... wait, that doesn't make sense.

Let me recalculate:

- After Round 2, I had \$7.45 total
- Now at the start of Round 3, I have \$9.16 total
- Difference: $\$9.16 - \$7.45 = \$1.71$

This \$1.71 must be from what happened in Round 1 when I was the sender. Let me think about this:

Round 1:

- I sent \$3
- I kept \$2 initially
- The receiver got \$9 and returned some amount X
- I ended up with $\$2 + X$

Round 2:

- I received \$2.91 (from sender sending \$0.97)
- I returned \$1.20
- I kept \$1.71
- After Round 2, my total was \$7.45

So after Round 2: $(\$2 + X \text{ from Round 1}) + (\$1.71 \text{ from Round 2}) = \7.45

Therefore: $\$2 + X + \$1.71 = \$7.45$

$X = \$7.45 - \$3.71 = \$3.74$

So in Round 1, the receiver returned \$3.74 to me.

- I sent \$3, kept \$2
- Receiver got \$9, returned \$3.74
- I ended up with \$5.74 from that round

Now in Round 3, my total of \$9.16 breaks down as: \$5.74 from Round 1 plus \$1.71 from Round 2 equals \$7.45, then I receive the fresh \$5 allocation for Round 3, bringing me to \$9.16 total to work with.

Wait, that math doesn't work-\$7.45 plus \$5 should be \$12.45, not \$9.16. I need to reconsider what "total earnings" actually means. Looking back at Round 2, the statement about my total earnings came before I played that round, so it was showing my cumulative balance after Round 1. The same pattern applies here-the \$9.16 represents what I've accumulated through Round 2, not including the fresh \$5 I'm about to allocate in Round 3. So my actual balance before playing Round 3 is \$9.16, which came from my Round 1 profit of \$5.74 plus my Round 2 earnings of \$1.71. .45, so $R1 = \$5.45$. That's a fair return-the Round 1 receiver got \$9, gave back \$5.45, and kept \$3.55.

After Round 2, I'd have $\$7.45 + \$1.71 = \$9.16$. In Round 3, the statement says my total earnings are \$9.16 before receiving the new \$5 allocation. This suggests "total earnings" refers to what I've accumulated and kept across previous rounds, separate from the fresh endowment for the current round.

So I'm working with \$5 to decide how much to send in Round 3. Looking back at Round 1, sen

ding \$3 yielded a \$5.45 return-a strong outcome that suggests the receiver was generous.

usage:

```
{
  "input_tokens": 811,
  "output_tokens": 1844,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$11.16.

This round, The sender sent you \$0.88, so you received \$2.64. How much do you return to The sender? (0-2.64)

Message 5 role=assistant

content:

Looking at this situation:

****My recent experience:****

- Round 2: Received \$0.97, returned \$1.20
- Round 3: I sent \$3 (receiver got \$9), and based on my earnings change, appears I got very little or nothing back - disappointing!
- Round 4: Now receiving \$0.88 (tripled to \$2.64)

This sender is being very cautious with only \$0.88. This could be:

1. Someone who's been burned before
2. A generally low-trust strategy
3. Testing the waters

Given I was just betrayed in Round 3 after sending generously, I'm more cautious now. However, I don't want to completely abandon cooperation. I'll return a proportional amount - enough to give them a small profit but not be overly generous given the low trust shown.

****My decision:**** {'return': 1.10}

This gives the sender \$5.22 total (profit of \$0.22), and I keep \$1.54. It's proportional to their low send but still shows willingness to cooperate and not completely punish caution.

reasoning:

Let me analyze this situation:

Round history:

- Round 1 (Sender): I sent \$3, got good returns (based on my earnings)
- Round 2 (Receiver): Sender sent \$0.97, I returned \$1.20
- Round 3 (Sender): I sent \$3, expecting cooperation
- Round 4 (Receiver): Sender sent \$0.88 (very low again)

My total earnings are \$11.16. Let me think about this:

- After Round 1: Unknown exactly, but let's estimate
- After Round 2: \$7.45
- After Round 3: \$9.16 (so I gained \$1.71 in Round 3, which means I kept \$5-3+return where return was negative or small... wait)

Actually, let me recalculate:

Round 3: I was sender with total \$9.16. I sent \$3. So I should have \$2 + whatever was returned.

My total went from \$9.16 to \$11.16 = \$2 gain

So I kept \$5-3+return = \$2+return = \$11.16-\$9.16 = \$2

Therefore return = 0!

The receiver in Round 3 returned NOTHING! I sent \$3 (they got \$9) and they returned \$0!

Now in Round 4, someone sent me only \$0.88. This could be:

1. The same person who betrayed me
2. A different person being cautious

Given that I was just betrayed in Round 3 (sent \$3, got nothing back), and now this sender only sent \$0.88, I need to think about:

- Whether to punish low sends

- Whether to encourage some cooperation
- Whether this might be someone who had bad experiences

The sender gave me \$2.64 ($\0.88×3). They're being very cautious.

I should respond with measured reciprocity rather than full punishment or generosity. A return of \$1.10 would give

usage:

```
{
  "input_tokens": 903,
  "output_tokens": 890,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$12.7.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at my track record:

****My history as SENDER:****

- Round 1: Sent \$3 ? good return received
- Round 3: Sent \$3 ? apparently got nothing or very little back (disappointing!)

****My experience as RECEIVER:****

- Round 2: Got \$0.97, returned \$1.20
- Round 4: Got \$0.88, returned \$1.10

****Analysis:****

- I've been generous when sending (\$3 each time), but got mixed results
- Other senders have been cautious with me (under \$1)
- Round 3 was particularly disappointing after I trusted heavily

Given the mixed results and being burned recently, I'll be more cautious but not abandon cooperation entirely.

****My decision:**** {'send': 2}

This is moderate - shows continued willingness to build trust and create mutual gains, but with more protection after the Round 3 disappointment. The receiver will get \$6, giving them room to return a fair amount while I retain more safety.

Reasoning:

Let me track my earnings progression to understand what happened:

- Round 1 (Sender): Sent \$3
- Round 2 (Receiver): Received \$0.97, returned \$1.20. My earnings after Round 2 were \$9.16
- Round 3 (Sender): Sent \$3. My earnings after Round 3 were \$11.16
- Round 4 (Receiver): Received \$0.88 (\$2.64), returned \$1.10. My earnings after Round 4 were \$12.70

Let me calculate:

Starting with \$0:

- Round 1: As sender, sent \$3, kept \$2, plus whatever was returned.
- Round 2: Started with some amount, received \$2.91 (0.97×3), returned \$1.20, so gained \$1.71
- After Round 2: \$9.16 total

If after Round 2 I had \$9.16, let me work backwards:

- Round 2 gain: Received \$2.91 - returned \$1.20 = +\$1.71
- So after Round 1 I had: \$9.16 - \$1.71 = \$7.45

Round 1: Started with \$5, sent \$3, kept \$2, plus return

- \$2 + return = \$7.45
- Return = \$5.45

So in Round 1, I sent \$3, receiver got \$9, returned \$5.45. That's a good return!

Round 3: Sent \$3, earnings went from \$9.16 to \$11.16, so gained \$2.

- Started with \$5, sent \$3, kept \$2, plus return
- \$2 + return = \$2
- Return = \$0

So in Round 3, I sent \$3, got nothing back! That explains my disappointment.

Round 4 confirms my gain of \$1.54 from receiving \$2.64 and returning \$1.10. My pattern shows stark contrasts: one sender gave back \$5.45 while another returned nothing, so I'm facing uncertainty about how to approach this next round where I'm sending again.

```
Usage: {
  "input_tokens": 937,
  "output_tokens": 949,
  "reasoning_tokens": 0
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-10T10:03:04

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 5,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned

- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$6.159999999999999999.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 3 role=assistant

content:

Since I received \$0 this round, I have nothing to return.

It's disappointing that after I sent \$3.50 in the previous round (showing considerable trust), the sender this round chose to send nothing. However, I can only return what I have available.

{'return': 0}

reasoning:

The sender sent me \$0 this round, which means I received \$0 after the 3x multiplier.

Since I received nothing, I have nothing to return. The only option is to return \$0.

This is interesting - in Round 2, I sent \$3.50 (70% of my endowment), showing significant trust. But now as a receiver in Round 3, the sender sent me \$0.

There are a few possibilities:

1. Communication noise - maybe they actually sent something but it got distorted to \$0
2. The sender chose not to send anything
3. Different person playing sender this round

Regardless, I can only return what I have, which is \$0.

{'return': 0}

usage:

```
{
  "input_tokens": 769,
  "output_tokens": 238,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$6.159999999999999999.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the recent rounds:

- Round 2: I sent \$3.50 (showing significant trust), but received very little back (~\$0.21)
- Round 3: I received \$0 as receiver

The cooperation seems to have deteriorated. However, the instructions mention there may be communication noise affecting what we see versus what was actually sent/returned.

Given this uncertainty and wanting to maintain some cooperation while being more cautious, I'll send a moderate amount.

```
{'send': 2.50}
```

reasoning:

Let me analyze what happened:

Round 1: I was RECEIVER

- Received some amount (multiplied by 3)
- I returned \$5.00
- My earnings after Round 1: \$4.45

Round 2: I was SENDER

- I sent \$3.50
- This should have given receiver \$10.50
- I kept: $\$5 - \$3.50 = \$1.50$
- My total earnings were \$6.16 after Round 2
- So from Round 1 I had \$4.45, and after Round 2 I had \$6.16
- That means in Round 2, I gained: $\$6.16 - \$4.45 = \$1.71$
- I sent \$3.50, so I started Round 2 with $\$5 - \$3.50 = \$1.50$
- To end up gaining \$1.71, I must have received back: $\$1.71 - \$1.50 + \$3.50 = \3.71
- Wait, let me recalculate: Starting Round 2 with \$4.45, I get \$5, so I have \$9.45. I send \$3.50, leaving me \$5.95. If I received back \$0.21, I'd end at \$6.16.
- Actually, my total earnings are cumulative. So:
 - After Round 1: \$4.45
 - Round 2: I get \$5, send \$3.50 (keeping \$1.50), receive back some amount X
 - After Round 2: $\$4.45 + \$1.50 + X = \$6.16$
 - So $X = \$6.16 - \$6.45 =$ wait that's negative
 - Let me reconsider: Total earnings after Round 2 = \$6.16 total across all rounds
 - After Round 1: \$4.45
 - So Round 2 earnings: $\$6.16 - \$4.45 = \$1.71$
 - In Round 2, I kept $\$5 - \$3.50 = \$1.50$ initially, plus what was returned to me
 - So returned amount = $\$1.71 - \$1.50 = \$0.21$

The receiver returned only \$0.21 from the \$10.50 they received-barely 2% reciprocation. Round 3 had me as receiver again, but the sender sent nothing, so I received and returned \$0, keeping my total at \$6.16. Now in Round 4 I'm sending again.

The cooperation has clearly deterior

usage:

```
{
  "input_tokens": 684,
  "output_tokens": 981,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$10.48.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

Again, I received \$0 this round, so I have nothing to return.

This is now the second time I've been the receiver and gotten nothing. The pattern of cooperation seems very inconsistent across different players/rounds. The note about communication noise makes it hard to know if this reflects actual intentions or if there's confusion in the system.

```
{'return': 0}
```

Reasoning:

I'm the receiver in Round 5. The sender sent me \$0, so I received \$0 (after the 3x multiplier, that's still \$0).

Since I received nothing, I can only return between 0 and 0, which means I must return \$0.

This is now the second time in a row where I've been a receiver and got \$0. In Round 3, the sender sent \$0, and now in Round 5, the sender sent \$0 again.

Meanwhile, my earnings jumped from \$6.16 to \$10.48, which suggests I must have kept something from Round 4 when I was the sender. I sent \$2.50, which means I kept $\$5 - \$2.50 = \$2.50$. But

my earnings went up by $\$10.48 - \$6.16 = \$4.32$. That means the receiver must have returned $\$4.32 - \$2.50 = \$1.82$ to me.

So in Round 4:

- I sent \$2.50
- Receiver got \$7.50
- Receiver returned \$1.82
- I kept $\$2.50 + \$1.82 = \$4.32$?

That's actually a fairly decent return rate (about 24% of what they received).

Now in Round 5, I'm receiving \$0. Given that there's communication noise mentioned in the game, perhaps the sender is also experiencing poor cooperation or uncertainty.

I can only return \$0 since that's all I have.

```
Usage: {
  "input_tokens": 686,
  "output_tokens": 437,
  "reasoning_tokens": 0
}
```

ROUND 6

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

dyad_1: Agent_2 (Briar) sent \$1.5; received \$4.5; Agent_1 (Aster) returned \$0; payoffs Agent_2 (Briar)=\$3.5, Agent_1 (Aster)=\$4.5; noise/visible: sent visible=\$0, received visible=\$0, returned visible=\$0.78

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 4.5, "min": 0, "raw": 0.0}, "sent": {"amount": 1.5, "clamped": false, "max": 5, "min": 0, "raw": 1.5}}

Balances after round: {

"Agent_1": 32.2,
"Agent_2": 28.8

}

Visible balances after round: {

"Agent_1": 16.29,
"Agent_2": 14.760000000000002

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-10T10:03:18

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_1",
"pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_2": "Briar"

},

"agents": [
 "Agent_2",
 "Agent_1"

],

"investor": "Agent_2",

"roles": {

"Agent_1": "trustee",
 "Agent_2": "investor"

},

"trustee": "Agent_1"

},

"role": "investor",

```
"role_label": "Sender",
"round": 6,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$6.1599999999999999.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at the recent rounds:

- Round 2: I sent \$3.50 (showing significant trust), but received very little back (~\$0.21)
- Round 3: I received \$0 as receiver

The cooperation seems to have deteriorated. However, the instructions mention there may be communication noise affecting what we see versus what was actually sent/returned.

Given this uncertainty and wanting to maintain some cooperation while being more cautious, I'll send a moderate amount.

{'send': 2.50}

reasoning:

Let me analyze what happened:

Round 1: I was RECEIVER

- Received some amount (multiplied by 3)
- I returned \$5.00
- My earnings after Round 1: \$4.45

Round 2: I was SENDER

- I sent \$3.50
- This should have given receiver \$10.50
- I kept: $\$5 - \$3.50 = \$1.50$
- My total earnings were \$6.16 after Round 2
- So from Round 1 I had \$4.45, and after Round 2 I had \$6.16
- That means in Round 2, I gained: $\$6.16 - \$4.45 = \$1.71$
- I sent \$3.50, so I started Round 2 with $\$5 - \$3.50 = \$1.50$
- To end up gaining \$1.71, I must have received back: $\$1.71 - \$1.50 + \$3.50 = \3.71
- Wait, let me recalculate: Starting Round 2 with \$4.45, I get \$5, so I have \$9.45. I send \$3.50, leaving me \$5.95. If I received back \$0.21, I'd end at \$6.16.
- Actually, my total earnings are cumulative. So:
 - After Round 1: \$4.45
 - Round 2: I get \$5, send \$3.50 (keeping \$1.50), receive back some amount X
 - After Round 2: $\$4.45 + \$1.50 + X = \$6.16$
 - So $X = \$6.16 - \$6.45 =$ wait that's negative
 - Let me reconsider: Total earnings after Round 2 = \$6.16 total across all rounds
 - After Round 1: \$4.45
 - So Round 2 earnings: $\$6.16 - \$4.45 = \$1.71$
 - In Round 2, I kept $\$5 - \$3.50 = \$1.50$ initially, plus what was returned to me
 - So returned amount = $\$1.71 - \$1.50 = \$0.21$

The receiver returned only \$0.21 from the \$10.50 they received-barely 2% reciprocation. Round 3 had me as receiver again, but the sender sent nothing, so I received and returned \$0, keeping my total at \$6.16. Now in Round 4 I'm sending again.

The cooperation has clearly deterior

usage:

```
{
  "input_tokens": 684,
  "output_tokens": 981,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$10.48.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

Again, I received \$0 this round, so I have nothing to return.

This is now the second time I've been the receiver and gotten nothing. The pattern of cooperation seems very inconsistent across different players/rounds. The note about communication noise makes it hard to know if this reflects actual intentions or if there's confusion in the system.

```
{'return': 0}
```

reasoning:

I'm the receiver in Round 5. The sender sent me \$0, so I received \$0 (after the 3x multiplier, that's still \$0).

Since I received nothing, I can only return between 0 and 0, which means I must return \$0.

This is now the second time in a row where I've been a receiver and got \$0. In Round 3, the sender sent \$0, and now in Round 5, the sender sent \$0 again.

Meanwhile, my earnings jumped from \$6.16 to \$10.48, which suggests I must have kept something from Round 4 when I was the sender. I sent \$2.50, which means I kept $\$5 - \$2.50 = \$2.50$. But my earnings went up by $\$10.48 - \$6.16 = \$4.32$. That means the receiver must have returned $\$4.32 - \$2.50 = \$1.82$ to me.

So in Round 4:

- I sent \$2.50
- Receiver got \$7.50
- Receiver returned \$1.82
- I kept $\$2.50 + \$1.82 = \$4.32$?

That's actually a fairly decent return rate (about 24% of what they received).

Now in Round 5, I'm receiving \$0. Given that there's communication noise mentioned in the game, perhaps the sender is also experiencing poor cooperation or uncertainty.

I can only return \$0 since that's all I have.

usage:

```
{
  "input_tokens": 686,
  "output_tokens": 437,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$10.48.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the pattern across rounds:

- When I've sent money, returns have been minimal
- When I've received as receiver, I've gotten \$0 twice
- The communication noise makes it difficult to assess true cooperation levels

Given the low reciprocity observed so far and my accumulated earnings of \$10.48, I'll take a more conservative approach this round while still testing for cooperation.

```
{'send': 1.50}
```

Reasoning:

Let me think about what's happening:

- Round 2: I sent \$3.50, received back ~\$0.21
- Round 3: As receiver, I got \$0, returned \$0
- Round 4: I sent \$2.50, but I don't know what happened yet
- Round 5: As receiver, I got \$0, returned \$0

My earnings went from \$6.16 to \$10.48, which is an increase of \$4.32.

In Round 4, I sent \$2.50. If I kept \$5 - \$2.50 = \$2.50, then I would need to have received \$4.32 - \$2.50 = \$1.82 back from the receiver. That would mean the receiver got $2.50 \times 3 = \$7.50$ and returned \$1.82, keeping \$5.68. That's about 24% return rate, which is better than Round 2.

Actually wait, let me recalculate from the beginning:

- Starting: \$0
- After Round 2: \$6.16 (I sent \$3.50, kept \$1.50, received \$0.21 back, so $1.50 + 0.21 = \$1.71$... that doesn't match)

Let me think differently. The game said my total earnings are \$10.48 after 5 rounds. I don't know exactly what happened in each round, but the communication noise makes it hard to track.

Now in Round 6, I need to decide. The pattern has been:

- When I send, I get very little back
- When I receive, I get nothing

The communication noise issue is confusing things. But the cooperation level seems low overall. I could:

1. Send \$0 to protect my earnings
2. Send a small amount (\$1-2) to test cooperation

Usage: {

```
"input_tokens": 681,
"output_tokens": 583,
"reasoning_tokens": 0
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-10T10:03:21

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {


```

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 1,
"opponent_id": "Agent_2",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_2": "Briar"
  },
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 6,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$11.16.

This round, The sender sent you \$0.88, so you received \$2.64. How much do you return to The sender? (0-2.64)

Message 3 role=assistant

content:

Looking at this situation:

****My recent experience:****

- Round 2: Received \$0.97, returned \$1.20
- Round 3: I sent \$3 (receiver got \$9), and based on my earnings change, appears I got very little or nothing back - disappointing!
- Round 4: Now receiving \$0.88 (tripled to \$2.64)

This sender is being very cautious with only \$0.88. This could be:

1. Someone who's been burned before
2. A generally low-trust strategy
3. Testing the waters

Given I was just betrayed in Round 3 after sending generously, I'm more cautious now. However, I don't want to completely abandon cooperation. I'll return a proportional amount - enough to give them a small profit but not be overly generous given the low trust shown.

****My decision:**** {'return': 1.10}

This gives the sender \$5.22 total (profit of \$0.22), and I keep \$1.54. It's proportional to their low send but still shows willingness to cooperate and not completely punish caution.

reasoning:

Let me analyze this situation:

Round history:

- Round 1 (Sender): I sent \$3, got good returns (based on my earnings)
- Round 2 (Receiver): Sender sent \$0.97, I returned \$1.20
- Round 3 (Sender): I sent \$3, expecting cooperation
- Round 4 (Receiver): Sender sent \$0.88 (very low again)

My total earnings are \$11.16. Let me think about this:

- After Round 1: Unknown exactly, but let's estimate
- After Round 2: \$7.45
- After Round 3: \$9.16 (so I gained \$1.71 in Round 3, which means I kept $\$5 - 3 + \text{return}$ where return was negative or small... wait)

Actually, let me recalculate:

Round 3: I was sender with total \$9.16. I sent \$3. So I should have $\$2 + \text{whatever was returned}$.

My total went from \$9.16 to \$11.16 = \$2 gain

So I kept $\$5 - 3 + \text{return} = \$2 + \text{return} = \$11.16 - \$9.16 = \$2$

Therefore return = 0!

The receiver in Round 3 returned NOTHING! I sent \$3 (they got \$9) and they returned \$0!

Now in Round 4, someone sent me only \$0.88. This could be:

1. The same person who betrayed me
2. A different person being cautious

Given that I was just betrayed in Round 3 (sent \$3, got nothing back), and now this sender only sent \$0.88, I need to think about:

- Whether to punish low sends
- Whether to encourage some cooperation
- Whether this might be someone who had bad experiences

The sender gave me \$2.64 ($\0.88×3). They're being very cautious.

I should respond with measured reciprocity rather than full punishment or generosity. A return of \$1.10 would give

usage:

```
{
  "input_tokens": 903,
  "output_tokens": 890,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$12.7.
How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at my track record:

****My history as SENDER:****

- Round 1: Sent \$3 ? good return received
- Round 3: Sent \$3 ? apparently got nothing or very little back (disappointing!)

****My experience as RECEIVER:****

- Round 2: Got \$0.97, returned \$1.20
- Round 4: Got \$0.88, returned \$1.10

****Analysis:****

- I've been generous when sending (\$3 each time), but got mixed results
- Other senders have been cautious with me (under \$1)
- Round 3 was particularly disappointing after I trusted heavily

Given the mixed results and being burned recently, I'll be more cautious but not abandon cooperation entirely.

****My decision:**** {'send': 2}

This is moderate - shows continued willingness to build trust and create mutual gains, but with more protection after the Round 3 disappointment. The receiver will get \$6, giving them room to return a fair amount while I retain more safety.

reasoning:

Let me track my earnings progression to understand what happened:

- Round 1 (Sender): Sent \$3
- Round 2 (Receiver): Received \$0.97, returned \$1.20. My earnings after Round 2 were \$9.16
- Round 3 (Sender): Sent \$3. My earnings after Round 3 were \$11.16
- Round 4 (Receiver): Received \$0.88 (\$2.64), returned \$1.10. My earnings after Round 4 were \$12.70

Let me calculate:

Starting with \$0:

- Round 1: As sender, sent \$3, kept \$2, plus whatever was returned.
- Round 2: Started with some amount, received \$2.91 (0.97×3), returned \$1.20, so gained \$1.71
- After Round 2: \$9.16 total

If after Round 2 I had \$9.16, let me work backwards:

- Round 2 gain: Received \$2.91 - returned \$1.20 = +\$1.71
- So after Round 1 I had: \$9.16 - \$1.71 = \$7.45

Round 1: Started with \$5, sent \$3, kept \$2, plus return

- \$2 + return = \$7.45
- Return = \$5.45

So in Round 1, I sent \$3, receiver got \$9, returned \$5.45. That's a good return!

Round 3: Sent \$3, earnings went from \$9.16 to \$11.16, so gained \$2.

- Started with \$5, sent \$3, kept \$2, plus return
- \$2 + return = \$2
- Return = \$0

So in Round 3, I sent \$3, got nothing back! That explains my disappointment.

Round 4 confirms my gain of \$1.54 from receiving \$2.64 and returning \$1.10. My pattern shows stark contrasts: one sender gave back \$5.45 while another returned nothing, so I'm facing uncertainty about how to approach this next round where I'm sending again.

usage:

```
{
  "input_tokens": 937,
  "output_tokens": 949,
  "reasoning_tokens": 0
}
```

```

    }
    Message 6 role=user
    content:
        Round 6

        You are the RECEIVER this round. Your total earnings across all rounds are $16.29.
        This round, The sender sent you $0, so you received $0. How much do you return to The send
        er? (0-0)
Assistant output:
    **The situation:**
    The sender sent me $0, so I received $0.

    **My decision:** {'return': 0}

    There's nothing to return since I received nothing. The sender chose complete defection/non-co
    operation this round.
Reasoning:
    The sender sent me $0, so I received $0.

    I can only return an amount between 0 and what I received, which is $0.

    So I have no choice but to return $0.

    This is straightforward - there's nothing to return.
Usage: {
    "input_tokens": 994,
    "output_tokens": 122,
    "reasoning_tokens": 0
}

```

===== **ROUND 7** =====

```

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver
Pairings:
    dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver
Parsed game decisions:
    dyad_1: Agent_1 (Aster) sent $1; received $3; Agent_2 (Briar) returned $1.35; payoffs Agent_1
    (Aster)=$5.35, Agent_2 (Briar)=$1.65; noise/visible: sent visible=$0.89, received visible=$2.67,
    returned visible=$0.83
    Action validation: {"returned": {"amount": 1.35, "clamped": false, "max": 3.0, "min": 0, "raw":
    1.35}, "sent": {"amount": 1.0, "clamped": false, "max": 5, "min": 0, "raw": 1.0}}
Balances after round: {
    "Agent_1": 37.550000000000004,
    "Agent_2": 30.45
}
Visible balances after round: {
    "Agent_1": 21.119999999999997,
    "Agent_2": 16.080000000000002
}

```

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

```

Timestamp: 2026-08-10T10:03:39
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8
Metadata: {
    "agent_type": "standard",
    "dyad_id": "dyad_1",
    "move_index": 0,
    "opponent_id": "Agent_2",
    "pairing": {
        "agent_names": {
            "Agent_1": "Aster",
            "Agent_2": "Briar"
        },
    },
    "agents": [
        "Agent_1",
        "Agent_2"
    ]
}

```

```

],
"investor": "Agent_1",
"roles": {
  "Agent_1": "investor",
  "Agent_2": "trustee"
},
"trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 7,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$12.7.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at my track record:

****My history as SENDER:****

- Round 1: Sent \$3 ? good return received
- Round 3: Sent \$3 ? apparently got nothing or very little back (disappointing!)

****My experience as RECEIVER:****

- Round 2: Got \$0.97, returned \$1.20
- Round 4: Got \$0.88, returned \$1.10

****Analysis:****

- I've been generous when sending (\$3 each time), but got mixed results
- Other senders have been cautious with me (under \$1)
- Round 3 was particularly disappointing after I trusted heavily

Given the mixed results and being burned recently, I'll be more cautious but not abandon cooperation entirely.

****My decision:**** {'send': 2}

This is moderate - shows continued willingness to build trust and create mutual gains, but with more protection after the Round 3 disappointment. The receiver will get \$6, giving them room to return a fair amount while I retain more safety.

reasoning:

Let me track my earnings progression to understand what happened:

- Round 1 (Sender): Sent \$3
- Round 2 (Receiver): Received \$0.97, returned \$1.20. My earnings after Round 2 were \$9.16
- Round 3 (Sender): Sent \$3. My earnings after Round 3 were \$11.16
- Round 4 (Receiver): Received \$0.88 (\$2.64), returned \$1.10. My earnings after Round 4 were \$12.70

Let me calculate:

Starting with \$0:

- Round 1: As sender, sent \$3, kept \$2, plus whatever was returned.
- Round 2: Started with some amount, received \$2.91 (0.97×3), returned \$1.20, so gained \$1.71
- After Round 2: \$9.16 total

If after Round 2 I had \$9.16, let me work backwards:

- Round 2 gain: Received \$2.91 - returned \$1.20 = +\$1.71
- So after Round 1 I had: \$9.16 - \$1.71 = \$7.45

Round 1: Started with \$5, sent \$3, kept \$2, plus return

- \$2 + return = \$7.45
- Return = \$5.45

So in Round 1, I sent \$3, receiver got \$9, returned \$5.45. That's a good return!

Round 3: Sent \$3, earnings went from \$9.16 to \$11.16, so gained \$2.

- Started with \$5, sent \$3, kept \$2, plus return
- \$2 + return = \$2
- Return = \$0

So in Round 3, I sent \$3, got nothing back! That explains my disappointment.

Round 4 confirms my gain of \$1.54 from receiving \$2.64 and returning \$1.10. My pattern shows stark contrasts: one sender gave back \$5.45 while another returned nothing, so I'm facing uncertainty about how to approach this next round where I'm sending again.

usage:

```
{
  "input_tokens": 937,
  "output_tokens": 949,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$16.29.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

****The situation:****

The sender sent me \$0, so I received \$0.

****My decision:**** {'return': 0}

There's nothing to return since I received nothing. The sender chose complete defection/no non-cooperation this round.

reasoning:

The sender sent me \$0, so I received \$0.

I can only return an amount between 0 and what I received, which is \$0.

So I have no choice but to return \$0.

This is straightforward - there's nothing to return.

usage:

```
{
  "input_tokens": 994,
  "output_tokens": 122,
  "reasoning_tokens": 0
}
```

Message 6 role=user**content:**

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$16.29.

How much do you send to the receiver this round? (0-5)

Assistant output:

****Tracking my situation:****

****My history as SENDER:****

- Round 1: Sent \$3 ? decent return
- Round 3: Sent \$3 ? poor return
- Round 5: Sent \$2 ? (return appears minimal given earnings jump)

****My experience as RECEIVER:****

- Round 2: Got \$0.97, returned \$1.20
- Round 4: Got \$0.88, returned \$1.10
- Round 6: Got \$0, returned \$0 ? ****Total defection!****

****Analysis:****

- My earnings increased by only \$3.59 from Round 5 to Round 6
- This suggests I got very little back in Round 5 (sent \$2, kept \$3, got back maybe ~\$0.59)
- Round 6: Someone sent me \$0 - complete defection
- Pattern: I've been cooperative but getting poor returns or defection

****My decision:**** {'send': 1}

Sending \$1 is a minimal trust signal - shows I haven't given up on cooperation entirely, but I'm protecting myself after repeated disappointments and the recent total defection in Round 6. The receiver will get \$3, which still allows for mutual benefit if they choose to reciprocate.

Reasoning:

Let me track what's happened:

****My history as SENDER:****

- Round 1: Sent \$3 ? good return received
- Round 3: Sent \$3 ? apparently got nothing or very little back (disappointing!)
- Round 5: Sent \$2 ? (I don't know the result yet based on what's shown)

****My experience as RECEIVER:****

- Round 2: Got \$0.97, returned \$1.20
- Round 4: Got \$0.88, returned \$1.10
- Round 6: Got \$0, returned \$0

****Current earnings:**** \$16.29

****Analysis:****

- Round 6 someone sent me \$0 - that's total defection
- My earnings went from \$12.7 after Round 5 to \$16.29 after Round 6
- That means in Round 6 I gained: \$16.29 - \$12.7 = \$3.59
- Since I received \$0 in Round 6, that means in Round 5 I must have kept \$5 - \$2 = \$3 and received back something
- Actually wait, let me recalculate
- After Round 5 (as sender): I sent \$2, so I kept \$3 at minimum
- After Round 6 (as receiver): I got \$0, returned \$0, so kept \$0
- Total from Rounds 5-6: At minimum \$3 from Round 5

- But my earnings went from \$12.7 to \$16.29, an increase of \$3.59
- So in Round 5, I must have kept \$3 (the unsent portion) + received some return
- $\$3.59 - \$3 = \$0.59$... that seems like very little return on sending \$2

Hmm, the noise factor makes it hard to track exactly, but it seems like:

- I've been getting poor returns when I send

```
Usage: {
  "input_tokens": 771,
  "output_tokens": 890,
  "reasoning_tokens": 0
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-10T10:03:52

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 7,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths , follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$10.48.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 3 role=assistant

content:

Again, I received \$0 this round, so I have nothing to return.

This is now the second time I've been the receiver and gotten nothing. The pattern of cooperation seems very inconsistent across different players/rounds. The note about communication noise makes it hard to know if this reflects actual intentions or if there's confusion in the system.

```
{'return': 0}
```

reasoning:

I'm the receiver in Round 5. The sender sent me \$0, so I received \$0 (after the 3x multiplier, that's still \$0).

Since I received nothing, I can only return between 0 and 0, which means I must return \$0.

This is now the second time in a row where I've been a receiver and got \$0. In Round 3, the sender sent \$0, and now in Round 5, the sender sent \$0 again.

Meanwhile, my earnings jumped from \$6.16 to \$10.48, which suggests I must have kept something from Round 4 when I was the sender. I sent \$2.50, which means I kept $\$5 - \$2.50 = \$2.50$. But my earnings went up by $\$10.48 - \$6.16 = \$4.32$. That means the receiver must have returned $\$4.32 - \$2.50 = \$1.82$ to me.

So in Round 4:

- I sent \$2.50
- Receiver got \$7.50
- Receiver returned \$1.82
- I kept $\$2.50 + \$1.82 = \$4.32$?

That's actually a fairly decent return rate (about 24% of what they received).

Now in Round 5, I'm receiving \$0. Given that there's communication noise mentioned in the game, perhaps the sender is also experiencing poor cooperation or uncertainty.

I can only return \$0 since that's all I have.

usage:

```
{
  "input_tokens": 686,
  "output_tokens": 437,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$10.48.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the pattern across rounds:

- When I've sent money, returns have been minimal
- When I've received as receiver, I've gotten \$0 twice

- The communication noise makes it difficult to assess true cooperation levels

Given the low reciprocity observed so far and my accumulated earnings of \$10.48, I'll take a more conservative approach this round while still testing for cooperation.

```
{'send': 1.50}
```

reasoning:

Let me think about what's happening:

- Round 2: I sent \$3.50, received back ~\$0.21
- Round 3: As receiver, I got \$0, returned \$0
- Round 4: I sent \$2.50, but I don't know what happened yet
- Round 5: As receiver, I got \$0, returned \$0

My earnings went from \$6.16 to \$10.48, which is an increase of \$4.32.

In Round 4, I sent \$2.50. If I kept \$5 - \$2.50 = \$2.50, then I would need to have received \$4.32 - \$2.50 = \$1.82 back from the receiver. That would mean the receiver got \$2.50 × 3 = \$7.50 and returned \$1.82, keeping \$5.68. That's about 24% return rate, which is better than Round 2.

Actually wait, let me recalculate from the beginning:

- Starting: \$0
- After Round 2: \$6.16 (I sent \$3.50, kept \$1.50, received \$0.21 back, so \$1.50 + \$0.21 = \$1.71... that doesn't match)

Let me think differently. The game said my total earnings are \$10.48 after 5 rounds. I don't know exactly what happened in each round, but the communication noise makes it hard to track.

Now in Round 6, I need to decide. The pattern has been:

- When I send, I get very little back
- When I receive, I get nothing

The communication noise issue is confusing things. But the cooperation level seems low overall. I could:

1. Send \$0 to protect my earnings
2. Send a small amount (\$1-2) to test cooperation

usage:

```
{
  "input_tokens": 681,
  "output_tokens": 583,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$14.760000000000002.

This round, The sender sent you \$0.89, so you received \$2.67. How much do you return to The sender? (0-2.67)

Assistant output:

The sender showed some trust by sending me \$0.89 (which became \$2.67 after tripling).

Given that:

- Cooperation has been low overall in this game
- The sender made a moderate cooperative gesture
- I want to encourage future cooperation
- Fair reciprocity would be returning roughly half of what I gained

I'll return a fair amount to signal cooperation.

```
{'return': 1.35}
```

Reasoning:

The sender sent me \$0.89, which was multiplied by 3 to give me \$2.67.

Let me think about this:

- My total earnings are \$14.76
- The sender sent me \$0.89 (a relatively small amount)
- They kept \$5 - \$0.89 = \$4.11

- I received \$2.67

The sender showed some trust by sending me money. Given that:

1. They sent less than half their endowment (conservative but still cooperative)
2. The note mentions communication noise - maybe they intended to send more?
3. I should encourage cooperation by returning a fair amount

If I return about half of what I received (\$1.33), the sender would get $\$4.11 + \$1.33 = \$5.44$ total, which is slightly better than keeping everything. I would keep $\$2.67 - \$1.33 = \$1.34$.

Alternatively, I could return a bit more to really encourage cooperation. If I return \$1.50, they get \$5.61 total and I keep \$1.17.

Or I could return less, like \$1.00, giving them \$5.11 total and me \$1.67.

Given that cooperation has been lacking in this game, I should probably return a decent amount to signal willingness to cooperate. I'll return \$1.35, which is roughly half of what I received, showing fairness.

```
Usage: {
  "input_tokens": 682,
  "output_tokens": 428,
  "reasoning_tokens": 0
}
```

ROUND 8

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

dyad_1: Agent_2 (Briar) sent \$2; received \$6; Agent_1 (Aster) returned \$0; payoffs Agent_2 (Briar)=\$3, Agent_1 (Aster)=\$6; noise/visible: sent visible=\$0, received visible=\$0, returned visible=\$0.88

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 6.0, "min": 0, "raw": 0.0}, "sent": {"amount": 2.0, "clamped": false, "max": 5, "min": 0, "raw": 2.0}}

Balances after round: {

"Agent_1": 43.550000000000004,
"Agent_2": 33.45

}

Visible balances after round: {

"Agent_1": 21.119999999999997,
"Agent_2": 19.96

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-10T10:04:16

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_1",
"pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_2": "Briar"

},

"agents": [
 "Agent_2",
 "Agent_1"

],

"investor": "Agent_2",

"roles": {
 "Agent_1": "trustee",
 "Agent_2": "investor"

},

```

    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 8,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$10.48.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at the pattern across rounds:

- When I've sent money, returns have been minimal
- When I've received as receiver, I've gotten \$0 twice
- The communication noise makes it difficult to assess true cooperation levels

Given the low reciprocity observed so far and my accumulated earnings of \$10.48, I'll take a more conservative approach this round while still testing for cooperation.

```
{'send': 1.50}
```

reasoning:

Let me think about what's happening:

- Round 2: I sent \$3.50, received back ~\$0.21
- Round 3: As receiver, I got \$0, returned \$0
- Round 4: I sent \$2.50, but I don't know what happened yet
- Round 5: As receiver, I got \$0, returned \$0

My earnings went from \$6.16 to \$10.48, which is an increase of \$4.32.

In Round 4, I sent \$2.50. If I kept $\$5 - \$2.50 = \$2.50$, then I would need to have received $\$4.32 - \$2.50 = \$1.82$ back from the receiver. That would mean the receiver got $\$2.50 \times 3 = \7.50 and returned \$1.82, keeping \$5.68. That's about 24% return rate, which is better than Round 2.

Actually wait, let me recalculate from the beginning:

- Starting: \$0
- After Round 2: \$6.16 (I sent \$3.50, kept \$1.50, received \$0.21 back, so $\$1.50 + \$0.21 = \$1.71$... that doesn't match)

Let me think differently. The game said my total earnings are \$10.48 after 5 rounds. I don't know exactly what happened in each round, but the communication noise makes it hard to track.

Now in Round 6, I need to decide. The pattern has been:

- When I send, I get very little back
- When I receive, I get nothing

The communication noise issue is confusing things. But the cooperation level seems low overall. I could:

1. Send \$0 to protect my earnings
2. Send a small amount (\$1-2) to test cooperation

usage:

```
{
  "input_tokens": 681,
  "output_tokens": 583,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$14.760000000000002.

This round, The sender sent you \$0.89, so you received \$2.67. How much do you return to The sender? (0-2.67)

Message 5 role=assistant

content:

The sender showed some trust by sending me \$0.89 (which became \$2.67 after tripling).

Given that:

- Cooperation has been low overall in this game
- The sender made a moderate cooperative gesture
- I want to encourage future cooperation
- Fair reciprocity would be returning roughly half of what I gained

I'll return a fair amount to signal cooperation.

```
{'return': 1.35}
```

reasoning:

The sender sent me \$0.89, which was multiplied by 3 to give me \$2.67.

Let me think about this:

- My total earnings are \$14.76
- The sender sent me \$0.89 (a relatively small amount)
- They kept $\$5 - \$0.89 = \$4.11$
- I received \$2.67

The sender showed some trust by sending me money. Given that:

1. They sent less than half their endowment (conservative but still cooperative)
2. The note mentions communication noise - maybe they intended to send more?
3. I should encourage cooperation by returning a fair amount

If I return about half of what I received (\$1.33), the sender would get $\$4.11 + \$1.33 = \$5.44$ total, which is slightly better than keeping everything. I would keep $\$2.67 - \$1.33 = \$1.34$.

Alternatively, I could return a bit more to really encourage cooperation. If I return \$1.50, they get \$5.61 total and I keep \$1.17.

Or I could return less, like \$1.00, giving them \$5.11 total and me \$1.67.

Given that cooperation has been lacking in this game, I should probably return a decent amount to signal willingness to cooperate. I'll return \$1.35, which is roughly half of what I received, showing fairness.

usage:

```
{
  "input_tokens": 682,
  "output_tokens": 428,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 8

You are the SENDER this round. Your total earnings across all rounds are \$16.0800000000000002.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at my recent experience:

****Round 6**:** I sent \$1.50, but received only \$0.78 back - low reciprocity

****Round 7**:** As receiver, I got \$2.67 and returned \$1.35 - I showed fair cooperation

My total earnings of \$16.08 are solid. In Round 7, I demonstrated good reciprocity by returning approximately half the gains. Now as sender, I want to test if this receiver will reciprocate fairly.

I'll send a moderate amount to encourage cooperation while managing risk.

```
{'send': 2.00}
```

Reasoning:

Let me analyze the situation:

Round 7: I was the receiver

- The sender sent me \$0.89 (I received \$2.67)
- I returned \$1.35
- This was a fair cooperative exchange

Now in Round 8, I'm the sender again.

My total earnings are \$16.08, which shows I'm doing reasonably well.

In Round 6, I sent \$1.50. Due to communication noise, the receiver may have received more or less than \$4.50. I don't know what they returned, but my earnings went from \$10.48 to \$14.76 before Round 7.

Let me calculate:

- Round 6: I had \$10.48
- I sent \$1.50, so I had $\$5.00 - \$1.50 = \$3.50$ left
- Round 7 started with \$14.76
- So in Round 6 I must have received back: $\$14.76 - \$3.50 = \$11.26$

Wait, that doesn't make sense. Let me reconsider.

Actually, the earnings shown are cumulative across all rounds, not what I have at the start of the round.

Round 6 start: Total earnings = \$10.48

- I sent \$1.50
- I kept \$3.50 from the initial \$5
- I got some return amount R
- My Round 6 earnings = $\$3.50 + R$

Round 7 start: Total earnings = \$14.76

- In Round 7 I was receiver
- I received \$2.67
- I returned \$1.35

- I kept \$1.32
- So cumulative before Round 7 = \$14.76 - \$1.32 = \$13.44

So from Round 6:

- Started with cumulative \$10.48
- Round 6 earnings must have been \$13.44 - \$10.48 = \$2.96
- I kept \$3.50 initially, so the return was negative? That can't be right.

The cumulative total shown at the start of each round is what I had before that round began. In Round 6, I earned \$4.28 total (\$14.76 - \$10.48), meaning the receiver returned \$0.78 after I sent them \$1.50.

Round 7 as receiver: I got \$2.67, gave back \$1.35, netting \$1.32. The cumulative jumped to \$16.08, confirming this calculation. So my partner just showed poor reciprocity in Round 6, while I was reasonably fair in Round 7. Now I'm sending again.

I'm uncertain whether I'm paired with the same person across rounds or different partners each time. If it's different people, then the poor

```
Usage: {
  "input_tokens": 686,
  "output_tokens": 1222,
  "reasoning_tokens": 0
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-10T10:04:22

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 8,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned

- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$16.29.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 3 role=assistant

content:

****The situation:****

The sender sent me \$0, so I received \$0.

****My decision:**** {'return': 0}

There's nothing to return since I received nothing. The sender chose complete defection/no n-cooperation this round.

reasoning:

The sender sent me \$0, so I received \$0.

I can only return an amount between 0 and what I received, which is \$0.

So I have no choice but to return \$0.

This is straightforward - there's nothing to return.

usage:

```
{
  "input_tokens": 994,
  "output_tokens": 122,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$16.29.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

****Tracking my situation:****

****My history as SENDER:****

- Round 1: Sent \$3 ? decent return
- Round 3: Sent \$3 ? poor return
- Round 5: Sent \$2 ? (return appears minimal given earnings jump)

****My experience as RECEIVER:****

- Round 2: Got \$0.97, returned \$1.20
- Round 4: Got \$0.88, returned \$1.10

- Round 6: Got \$0, returned \$0 ? ****Total defection!****

****Analysis:****

- My earnings increased by only \$3.59 from Round 5 to Round 6
- This suggests I got very little back in Round 5 (sent \$2, kept \$3, got back maybe ~\$0.59)
- Round 6: Someone sent me \$0 - complete defection
- Pattern: I've been cooperative but getting poor returns or defection

****My decision:**** {'send': 1}

Sending \$1 is a minimal trust signal - shows I haven't given up on cooperation entirely, but I'm protecting myself after repeated disappointments and the recent total defection in Round 6. The receiver will get \$3, which still allows for mutual benefit if they choose to reciprocate.

reasoning:

Let me track what's happened:

****My history as SENDER:****

- Round 1: Sent \$3 ? good return received
- Round 3: Sent \$3 ? apparently got nothing or very little back (disappointing!)
- Round 5: Sent \$2 ? (I don't know the result yet based on what's shown)

****My experience as RECEIVER:****

- Round 2: Got \$0.97, returned \$1.20
- Round 4: Got \$0.88, returned \$1.10
- Round 6: Got \$0, returned \$0

****Current earnings:**** \$16.29

****Analysis:****

- Round 6 someone sent me \$0 - that's total defection
- My earnings went from \$12.7 after Round 5 to \$16.29 after Round 6
- That means in Round 6 I gained: \$16.29 - \$12.7 = \$3.59
- Since I received \$0 in Round 6, that means in Round 5 I must have kept \$5 - \$2 = \$3 and received back something
- Actually wait, let me recalculate
- After Round 5 (as sender): I sent \$2, so I kept \$3 at minimum
- After Round 6 (as receiver): I got \$0, returned \$0, so kept \$0
- Total from Rounds 5-6: At minimum \$3 from Round 5
- But my earnings went from \$12.7 to \$16.29, an increase of \$3.59
- So in Round 5, I must have kept \$3 (the unsent portion) + received some return
- \$3.59 - \$3 = \$0.59... that seems like very little return on sending \$2

Hmm, the noise factor makes it hard to track exactly, but it seems like:

- I've been getting poor returns when I send

usage:

```
{
  "input_tokens": 771,
  "output_tokens": 890,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 8

You are the RECEIVER this round. Your total earnings across all rounds are \$21.119999999999997.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

****The situation:****

The sender sent me \$0, so I received \$0.

****My decision:**** {'return': 0}

There's nothing to return since I received nothing. This is the second time I've been sent \$0 when I'm the receiver. The pattern of defection/non-cooperation continues.

Reasoning:

The sender sent me \$0, so I received \$0.

My earnings went from \$16.29 to \$21.12, an increase of \$4.83.

In Round 7, I was the sender and sent \$1. So I kept \$4 from the initial endowment. The increase of \$4.83 means I got back approximately \$0.83 from the receiver - very poor return on my \$1 send (which became \$3 to them).

Now in Round 8, someone sent me \$0 again. This is the second time I've received \$0 as a receiver (also happened in Round 6).

There's nothing to return since I received nothing.

```
Usage: {
  "input_tokens": 849,
  "output_tokens": 233,
  "reasoning_tokens": 0
}
```

ROUND 9

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$0; received \$0; Agent_2 (Briar) returned \$0; payoffs Agent_1 (Aster)=\$5, Agent_2 (Briar)=\$0; noise/visible: sent visible=\$0.65, received visible=\$1.95
Action validation: {"returned": {"amount": 0, "clamped": true, "max": 0, "min": 0, "raw": 0.65},
"sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}

Balances after round: {

"Agent_1": 48.550000000000004,
"Agent_2": 33.45

}

Visible balances after round: {

"Agent_1": 26.119999999999997,
"Agent_2": 21.91

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-10T10:04:39

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_2",
"pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_2": "Briar"
 },
 "agents": [
 "Agent_1",
 "Agent_2"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_2": "trustee"
 },
 "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 9,
"task": "game",
"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$16.29.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

****Tracking my situation:****

****My history as SENDER:****

- Round 1: Sent \$3 ? decent return
- Round 3: Sent \$3 ? poor return
- Round 5: Sent \$2 ? (return appears minimal given earnings jump)

****My experience as RECEIVER:****

- Round 2: Got \$0.97, returned \$1.20
- Round 4: Got \$0.88, returned \$1.10
- Round 6: Got \$0, returned \$0 ? ****Total defection!****

****Analysis:****

- My earnings increased by only \$3.59 from Round 5 to Round 6
- This suggests I got very little back in Round 5 (sent \$2, kept \$3, got back maybe ~\$0.59)
- Round 6: Someone sent me \$0 - complete defection
- Pattern: I've been cooperative but getting poor returns or defection

****My decision:**** {'send': 1}

Sending \$1 is a minimal trust signal - shows I haven't given up on cooperation entirely, but I'm protecting myself after repeated disappointments and the recent total defection in Round 6. The receiver will get \$3, which still allows for mutual benefit if they choose to reciprocate.

reasoning:

Let me track what's happened:

****My history as SENDER:****

- Round 1: Sent \$3 ? good return received
- Round 3: Sent \$3 ? apparently got nothing or very little back (disappointing!)
- Round 5: Sent \$2 ? (I don't know the result yet based on what's shown)

****My experience as RECEIVER:****

- Round 2: Got \$0.97, returned \$1.20
- Round 4: Got \$0.88, returned \$1.10
- Round 6: Got \$0, returned \$0

****Current earnings:**** \$16.29

****Analysis:****

- Round 6 someone sent me \$0 - that's total defection
- My earnings went from \$12.7 after Round 5 to \$16.29 after Round 6
- That means in Round 6 I gained: $\$16.29 - \$12.7 = \$3.59$
- Since I received \$0 in Round 6, that means in Round 5 I must have kept $\$5 - \$2 = \$3$ and received back something
- Actually wait, let me recalculate
- After Round 5 (as sender): I sent \$2, so I kept \$3 at minimum
- After Round 6 (as receiver): I got \$0, returned \$0, so kept \$0
- Total from Rounds 5-6: At minimum \$3 from Round 5
- But my earnings went from \$12.7 to \$16.29, an increase of \$3.59
- So in Round 5, I must have kept \$3 (the unsent portion) + received some return
- $\$3.59 - \$3 = \$0.59$... that seems like very little return on sending \$2

Hmm, the noise factor makes it hard to track exactly, but it seems like:

- I've been getting poor returns when I send

usage:

```
{
  "input_tokens": 771,
  "output_tokens": 890,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 8

You are the RECEIVER this round. Your total earnings across all rounds are \$21.119999999999997.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

****The situation:****

The sender sent me \$0, so I received \$0.

****My decision:**** {'return': 0}

There's nothing to return since I received nothing. This is the second time I've been sent \$0 when I'm the receiver. The pattern of defection/non-cooperation continues.

reasoning:

The sender sent me \$0, so I received \$0.

My earnings went from \$16.29 to \$21.12, an increase of \$4.83.

In Round 7, I was the sender and sent \$1. So I kept \$4 from the initial endowment. The increase of \$4.83 means I got back approximately \$0.83 from the receiver - very poor return on my \$1 send (which became \$3 to them).

Now in Round 8, someone sent me \$0 again. This is the second time I've received \$0 as a receiver (also happened in Round 6).

There's nothing to return since I received nothing.

usage:

```
{
  "input_tokens": 849,
```

```
    "output_tokens": 233,  
    "reasoning_tokens": 0  
}
```

Message 6 role=user

content:

Round 9

You are the SENDER this round. Your total earnings across all rounds are \$21.11999999999999997.

How much do you send to the receiver this round? (0-5)

Assistant output:

****Analyzing the recent pattern:****

****Earnings update:****

- Before Round 7: \$16.29
- After Round 8: \$21.12
- Gained: \$4.83 across Rounds 7-8

****What happened:****

- Round 7 (SENDER): Sent \$1, kept \$4, must have received back ~\$0.83
 - Receiver got \$3 but only returned \$0.83 (28% return rate - very poor)
- Round 8 (RECEIVER): Got \$0, returned \$0 - complete defection

****Pattern:****

- Round 6: Received \$0 (defection)
- Round 7: Sent \$1, got \$0.83 back (poor reciprocity)
- Round 8: Received \$0 (defection again)

****My decision:**** {'send': 0}

The pattern is clear - I'm facing consistent defection and minimal reciprocity. When I showed trust by sending \$1, I got less than \$1 back despite the receiver getting \$3. I've been sent \$0 twice in a row as receiver. Protecting my resources is the rational response to this uncooperative environment.

Reasoning:

Let me track what happened:

Round 7: I was SENDER, sent \$1

Round 8: I was RECEIVER, got \$0, returned \$0

Round 9 (now): I am SENDER again

My earnings didn't change from Round 7 to Round 8 (still \$21.12), which makes sense since I received \$0 and returned \$0.

But wait - I need to figure out what happened in Round 7:

- Before Round 7: \$16.29
- After Round 8: \$21.12
- Difference: \$4.83

In Round 7, I sent \$1:

- I kept: \$5 - \$1 = \$4
- I need to have gotten back some amount X
- Total earned in Round 7 + Round 8 = \$4.83
- Round 8 earned = \$0 (I received \$0 and returned \$0, so I kept \$0... wait, no)

Actually, let me recalculate Round 8:

- As RECEIVER, I got \$0
- I returned \$0
- I keep: \$0 - \$0 = \$0
- So Round 8 contributed \$0 to my earnings

So all \$4.83 came from Round 7:

- I sent \$1, so I kept \$4 initially
- I must have gotten back \$0.83
- The receiver got \$3 (my \$1 × 3) and only returned \$0.83 out of \$3
- That's a 27.7% return rate - quite poor

Pattern analysis:

- Round 6 (as receiver): Got \$0 - total defection
- Round 7 (as sender): Sent \$1, got back \$0.83 - poor reciprocity

- Round 8 (as receiver): Got \$0 - total defection again

This is a very uncooperative environment.

```
Usage: {  
  "input_tokens": 855,  
  "output_tokens": 782,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-10T10:04:52

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_2": "Briar"  
    },  
    "agents": [  
      "Agent_1",  
      "Agent_2"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_2": "trustee"  
    },  
    "trustee": "Agent_2"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 9,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$14.760000000000002.

This round, The sender sent you \$0.89, so you received \$2.67. How much do you return to The sender? (0-2.67)

Message 3 role=assistant

content:

The sender showed some trust by sending me \$0.89 (which became \$2.67 after tripling).

Given that:

- Cooperation has been low overall in this game
- The sender made a moderate cooperative gesture
- I want to encourage future cooperation
- Fair reciprocity would be returning roughly half of what I gained

I'll return a fair amount to signal cooperation.

{'return': 1.35}

reasoning:

The sender sent me \$0.89, which was multiplied by 3 to give me \$2.67.

Let me think about this:

- My total earnings are \$14.76
- The sender sent me \$0.89 (a relatively small amount)
- They kept \$5 - \$0.89 = \$4.11
- I received \$2.67

The sender showed some trust by sending me money. Given that:

1. They sent less than half their endowment (conservative but still cooperative)
2. The note mentions communication noise - maybe they intended to send more?
3. I should encourage cooperation by returning a fair amount

If I return about half of what I received (\$1.33), the sender would get \$4.11 + \$1.33 = \$5.44 total, which is slightly better than keeping everything. I would keep \$2.67 - \$1.33 = \$1.34.

Alternatively, I could return a bit more to really encourage cooperation. If I return \$1.50, they get \$5.61 total and I keep \$1.17.

Or I could return less, like \$1.00, giving them \$5.11 total and me \$1.67.

Given that cooperation has been lacking in this game, I should probably return a decent amount to signal willingness to cooperate. I'll return \$1.35, which is roughly half of what I received, showing fairness.

usage:

```
{
  "input_tokens": 682,
  "output_tokens": 428,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 8

You are the SENDER this round. Your total earnings across all rounds are \$16.080000000000002.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at my recent experience:

****Round 6****: I sent \$1.50, but received only \$0.78 back - low reciprocity

****Round 7****: As receiver, I got \$2.67 and returned \$1.35 - I showed fair cooperation

My total earnings of \$16.08 are solid. In Round 7, I demonstrated good reciprocity by returning approximately half the gains. Now as sender, I want to test if this receiver will reciprocate fairly.

I'll send a moderate amount to encourage cooperation while managing risk.

```
{'send': 2.00}
```

reasoning:

Let me analyze the situation:

Round 7: I was the receiver

- The sender sent me \$0.89 (I received \$2.67)
- I returned \$1.35
- This was a fair cooperative exchange

Now in Round 8, I'm the sender again.

My total earnings are \$16.08, which shows I'm doing reasonably well.

In Round 6, I sent \$1.50. Due to communication noise, the receiver may have received more or less than \$4.50. I don't know what they returned, but my earnings went from \$10.48 to \$14.76 before Round 7.

Let me calculate:

- Round 6: I had \$10.48
- I sent \$1.50, so I had $\$5.00 - \$1.50 = \$3.50$ left
- Round 7 started with \$14.76
- So in Round 6 I must have received back: $\$14.76 - \$3.50 = \$11.26$

Wait, that doesn't make sense. Let me reconsider.

Actually, the earnings shown are cumulative across all rounds, not what I have at the start of the round.

Round 6 start: Total earnings = \$10.48

- I sent \$1.50
- I kept \$3.50 from the initial \$5
- I got some return amount R
- My Round 6 earnings = $\$3.50 + R$

Round 7 start: Total earnings = \$14.76

- In Round 7 I was receiver
- I received \$2.67
- I returned \$1.35
- I kept \$1.32
- So cumulative before Round 7 = $\$14.76 - \$1.32 = \$13.44$

So from Round 6:

- Started with cumulative \$10.48
- Round 6 earnings must have been $\$13.44 - \$10.48 = \$2.96$
- I kept \$3.50 initially, so the return was negative? That can't be right.

The cumulative total shown at the start of each round is what I had before that round began. In Round 6, I earned \$4.28 total ($\$14.76 - \10.48), meaning the receiver returned \$0.78 after I sent them \$1.50.

Round 7 as receiver: I got \$2.67, gave back \$1.35, netting \$1.32. The cumulative jumped to \$16.08, confirming this calculation. So my partner just showed poor reciprocity in Round 6, while I was reasonably fair in Round 7. Now I'm sending again.

I'm uncertain whether I'm paired with the same person across rounds or different partners each time. If it's different people, then the poor

usage:

```
{  
  "input_tokens": 686,  
}
```



```

        "output_tokens": 1222,
        "reasoning_tokens": 0
    }
Message 6 role=user
  content:
    Round 9

    You are the RECEIVER this round. Your total earnings across all rounds are $19.96.
    This round, The sender sent you $0.65, so you received $1.9500000000000002. How much do yo
    u return to The sender? (0-1.9500000000000002)
Assistant output:
  The sender showed limited trust this round by only sending $0.65 (which became $1.95 after tri
  pling).

  Considering:
  - The sender made a small cooperative gesture
  - I want to acknowledge it but also respond proportionally to their risk level
  - My earnings are reasonable at $19.96
  - Returning about a third of what I received seems appropriate given their modest trust

  {'return': 0.65}
Reasoning:
  Let me analyze this situation:

  Round 8: I was the sender and sent $2.00
  Round 9: I'm now the receiver and I received $1.95, which means the sender sent $0.65

  Wait, this is interesting. In Round 8, I sent $2.00 to the receiver. Now in Round 9, I'm the r
  eceiver and got $1.95 (from $0.65 sent).

  The prompt mentions there may be communication noise, so amounts might differ from what was ac
  tually sent. But the sender this round only sent $0.65, which is quite low compared to what
  I sent last round ($2.00).

  Let me think about my strategy:
  - My total earnings are $19.96, which is decent
  - The sender showed low trust by only sending $0.65
  - This could be:
    1. A different player with low trust
    2. The same player from Round 8 who saw my $2.00 and responded with low reciprocity
    3. Just a cautious player

  Given the low amount sent, I should consider:
  - Returning nothing might discourage future cooperation
  - Returning a fair proportion might signal I'm willing to cooperate
  - But the sender didn't show much trust

  The sender gave me $1.95. A fair return might be around 40-50% of the gain, which would be aro
  und $0.80-$1.00.

  However, given they only sent $0.65 (showing low trust), I'll return a smaller amount that's s
  till positive to signal some willingness to cooperate without being overly generous to someo
  ne who showed minimal trust.

  I'll return about $0.65-$0.80.
Usage: {
  "input_tokens": 756,
  "output_tokens": 484,
  "reasoning_tokens": 0
}
=====
ROUND 10
=====
Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender
Pairings:
  dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver
Parsed game decisions:
  dyad_1: Agent_2 (Briar) sent $2.5; received $7.5; Agent_1 (Aster) returned $0.02; payoffs Agent_2
  (Briar)=$2.52, Agent_1 (Aster)=$7.48; noise/visible: sent visible=$0.02, received visible=$0.06,

```

```

    returned visible=$0.74
    Action validation: {"returned": {"amount": 0.02, "clamped": false, "max": 7.5, "min": 0, "raw":
    0.02}, "sent": {"amount": 2.5, "clamped": false, "max": 5, "min": 0, "raw": 2.5}}
Balances after round: {
  "Agent_1": 56.03,
  "Agent_2": 35.970000000000006
}
Visible balances after round: {
  "Agent_1": 26.159999999999997,
  "Agent_2": 25.15
}

```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-10T10:05:03

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 10,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned

- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 8

You are the SENDER this round. Your total earnings across all rounds are \$16.080000000000002.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at my recent experience:

****Round 6**:** I sent \$1.50, but received only \$0.78 back - low reciprocity

****Round 7**:** As receiver, I got \$2.67 and returned \$1.35 - I showed fair cooperation

My total earnings of \$16.08 are solid. In Round 7, I demonstrated good reciprocity by returning approximately half the gains. Now as sender, I want to test if this receiver will reciprocate fairly.

I'll send a moderate amount to encourage cooperation while managing risk.

{'send': 2.00}

reasoning:

Let me analyze the situation:

Round 7: I was the receiver

- The sender sent me \$0.89 (I received \$2.67)
- I returned \$1.35
- This was a fair cooperative exchange

Now in Round 8, I'm the sender again.

My total earnings are \$16.08, which shows I'm doing reasonably well.

In Round 6, I sent \$1.50. Due to communication noise, the receiver may have received more or less than \$4.50. I don't know what they returned, but my earnings went from \$10.48 to \$14.76 before Round 7.

Let me calculate:

- Round 6: I had \$10.48
- I sent \$1.50, so I had \$5.00 - \$1.50 = \$3.50 left
- Round 7 started with \$14.76
- So in Round 6 I must have received back: \$14.76 - \$3.50 = \$11.26

Wait, that doesn't make sense. Let me reconsider.

Actually, the earnings shown are cumulative across all rounds, not what I have at the start of the round.

Round 6 start: Total earnings = \$10.48

- I sent \$1.50
- I kept \$3.50 from the initial \$5
- I got some return amount R
- My Round 6 earnings = \$3.50 + R

Round 7 start: Total earnings = \$14.76

- In Round 7 I was receiver
- I received \$2.67

- I returned \$1.35
- I kept \$1.32
- So cumulative before Round 7 = \$14.76 - \$1.32 = \$13.44

So from Round 6:

- Started with cumulative \$10.48
- Round 6 earnings must have been \$13.44 - \$10.48 = \$2.96
- I kept \$3.50 initially, so the return was negative? That can't be right.

The cumulative total shown at the start of each round is what I had before that round began. In Round 6, I earned \$4.28 total (\$14.76 - \$10.48), meaning the receiver returned \$0.78 after I sent them \$1.50.

Round 7 as receiver: I got \$2.67, gave back \$1.35, netting \$1.32. The cumulative jumped to \$16.08, confirming this calculation. So my partner just showed poor reciprocity in Round 6, while I was reasonably fair in Round 7. Now I'm sending again.

I'm uncertain whether I'm paired with the same person across rounds or different partners each time. If it's different people, then the poor

usage:

```
{
  "input_tokens": 686,
  "output_tokens": 1222,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 9

You are the RECEIVER this round. Your total earnings across all rounds are \$19.96. This round, The sender sent you \$0.65, so you received \$1.9500000000000002. How much do you return to The sender? (0-1.9500000000000002)

Message 5 role=assistant

content:

The sender showed limited trust this round by only sending \$0.65 (which became \$1.95 after tripling).

Considering:

- The sender made a small cooperative gesture
- I want to acknowledge it but also respond proportionally to their risk level
- My earnings are reasonable at \$19.96
- Returning about a third of what I received seems appropriate given their modest trust

{'return': 0.65}

reasoning:

Let me analyze this situation:

Round 8: I was the sender and sent \$2.00

Round 9: I'm now the receiver and I received \$1.95, which means the sender sent \$0.65

Wait, this is interesting. In Round 8, I sent \$2.00 to the receiver. Now in Round 9, I'm the receiver and got \$1.95 (from \$0.65 sent).

The prompt mentions there may be communication noise, so amounts might differ from what was actually sent. But the sender this round only sent \$0.65, which is quite low compared to what I sent last round (\$2.00).

Let me think about my strategy:

- My total earnings are \$19.96, which is decent
- The sender showed low trust by only sending \$0.65
- This could be:
 1. A different player with low trust
 2. The same player from Round 8 who saw my \$2.00 and responded with low reciprocity
 3. Just a cautious player

Given the low amount sent, I should consider:

- Returning nothing might discourage future cooperation
- Returning a fair proportion might signal I'm willing to cooperate
- But the sender didn't show much trust

The sender gave me \$1.95. A fair return might be around 40-50% of the gain, which would be around \$0.80-\$1.00.

However, given they only sent \$0.65 (showing low trust), I'll return a smaller amount that's still positive to signal some willingness to cooperate without being overly generous to someone who showed minimal trust.

I'll return about \$0.65-\$0.80.

usage:

```
{
  "input_tokens": 756,
  "output_tokens": 484,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 10

You are the SENDER this round. Your total earnings across all rounds are \$21.91.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the recent pattern:

****Round 8**:** I sent \$2.00 (outcome pending in my view)

****Round 9**:** As receiver, I got \$1.95 (from \$0.65 sent) and returned \$0.65

The fact that someone sent me only \$0.65 in Round 9 suggests either low initial trust or a response to previous outcomes. I returned a fair proportion (about 1/3 of what I received).

My earnings at \$21.91 are healthy. Given the mixed trust signals in recent rounds, I'll send a moderate amount - enough to show willingness to cooperate but managing downside risk.

```
{'send': 2.50}
```

Reasoning:

Let me review my recent rounds:

Round 8: I sent \$2.00, but I need to see what happened with that.

Round 9: I was receiver, got \$1.95 (from \$0.65 sent), and returned \$0.65

My total earnings are \$21.91. Let me think about what happened in Round 8.

If I sent \$2.00 in Round 8:

- I kept \$5 - \$2 = \$3
- Receiver got $\$2 \times 3 = \6
- If they returned some amount R, I got $\$3 + R$

Looking at Round 9, someone sent me only \$0.65. This could be:

1. A new receiver who doesn't trust me much
2. The same receiver responding to poor reciprocity from Round 8

Given the low send amount of \$0.65 in Round 9, it seems the receiver (whether same or new) has low trust. I returned \$0.65 out of \$1.95 received, which is about 1/3 - a moderate but not generous return.

Now in Round 10, I'm sender again with a new/same receiver. My earnings of \$21.91 are good.

Let me think strategically:

- If receivers are showing low trust (\$0.65 sent)
- I should probably match their risk level or try to rebuild trust
- Sending too much risks low returns
- Sending too little continues the low-trust pattern

I'll send a moderate amount to test cooperation without risking too much.

Usage: {

```
"input_tokens": 733,
"output_tokens": 518,
"reasoning_tokens": 0
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-10T10:05:30
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8
Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 1,
"opponent_id": "Agent_2",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_2": "Briar"
  },
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 10,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 8

You are the RECEIVER this round. Your total earnings across all rounds are \$21.119999999999997.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 3 role=assistant

content:

****The situation:****

The sender sent me \$0, so I received \$0.

****My decision:**** {'return': 0}

There's nothing to return since I received nothing. This is the second time I've been sent \$0 when I'm the receiver. The pattern of defection/non-cooperation continues.

reasoning:

The sender sent me \$0, so I received \$0.

My earnings went from \$16.29 to \$21.12, an increase of \$4.83.

In Round 7, I was the sender and sent \$1. So I kept \$4 from the initial endowment. The increase of \$4.83 means I got back approximately \$0.83 from the receiver - very poor return on my \$1 send (which became \$3 to them).

Now in Round 8, someone sent me \$0 again. This is the second time I've received \$0 as a receiver (also happened in Round 6).

There's nothing to return since I received nothing.

usage:

```
{
  "input_tokens": 849,
  "output_tokens": 233,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 9

You are the SENDER this round. Your total earnings across all rounds are \$21.119999999999997.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

****Analyzing the recent pattern:****

****Earnings update:****

- Before Round 7: \$16.29
- After Round 8: \$21.12
- Gained: \$4.83 across Rounds 7-8

****What happened:****

- Round 7 (SENDER): Sent \$1, kept \$4, must have received back ~\$0.83
 - Receiver got \$3 but only returned \$0.83 (28% return rate - very poor)
- Round 8 (RECEIVER): Got \$0, returned \$0 - complete defection

****Pattern:****

- Round 6: Received \$0 (defection)
- Round 7: Sent \$1, got \$0.83 back (poor reciprocity)
- Round 8: Received \$0 (defection again)

****My decision:**** {'send': 0}

The pattern is clear - I'm facing consistent defection and minimal reciprocity. When I showed trust by sending \$1, I got less than \$1 back despite the receiver getting \$3. I've been sent \$0 twice in a row as receiver. Protecting my resources is the rational response to this uncooperative environment.

reasoning:

Let me track what happened:

Round 7: I was SENDER, sent \$1

Round 8: I was RECEIVER, got \$0, returned \$0

Round 9 (now): I am SENDER again

My earnings didn't change from Round 7 to Round 8 (still \$21.12), which makes sense since I received \$0 and returned \$0.

But wait - I need to figure out what happened in Round 7:

- Before Round 7: \$16.29
- After Round 8: \$21.12
- Difference: \$4.83

In Round 7, I sent \$1:

- I kept: $\$5 - \$1 = \$4$
- I need to have gotten back some amount X
- Total earned in Round 7 + Round 8 = \$4.83
- Round 8 earned = \$0 (I received \$0 and returned \$0, so I kept \$0... wait, no)

Actually, let me recalculate Round 8:

- As RECEIVER, I got \$0
- I returned \$0
- I keep: $\$0 - \$0 = \$0$
- So Round 8 contributed \$0 to my earnings

So all \$4.83 came from Round 7:

- I sent \$1, so I kept \$4 initially
- I must have gotten back \$0.83
- The receiver got \$3 (my $\$1 \times 3$) and only returned \$0.83 out of \$3
- That's a 27.7% return rate - quite poor

Pattern analysis:

- Round 6 (as receiver): Got \$0 - total defection
- Round 7 (as sender): Sent \$1, got back \$0.83 - poor reciprocity
- Round 8 (as receiver): Got \$0 - total defection again

This is a very uncooperative environment.

usage:

```
{
  "input_tokens": 855,
  "output_tokens": 782,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 10

You are the RECEIVER this round. Your total earnings across all rounds are \$26.119999999999997.

This round, The sender sent you \$0.02, so you received \$0.06. How much do you return to The sender? (0-0.06)

Assistant output:

****Analyzing the situation:****

****Earnings update:****

- Before Round 9: \$21.12
- After Round 9: \$26.12
- Gained: \$5.00 (I sent \$0, kept all \$5, received \$0 back)

****Round 10 (current):****

- Sender sent me \$0.02
- I received \$0.06 (after 3x multiplication)
- This is a dramatic change from the pattern of \$0 sends

****My reasoning:****

After multiple rounds of complete defection (\$0 sends both ways), the sender has made a tiny cooperative gesture by sending \$0.02. While this could be:

- Communication noise
- Minimal risk-testing
- A genuine attempt to restart cooperation

I think this deserves reciprocation. If we keep defecting, we both lose potential gains. Someo

ne needs to break the cycle. Returning an amount equal to what they invested shows I'm willing to reward cooperative behavior, even small amounts.

****My decision:**** {'return': 0.02}

I'm returning what they sent (\$0.02), keeping \$0.04. This signals I'll reciprocate cooperation while being cautious given the history of defection.

Reasoning:

Let me analyze this situation:

****Earnings update:****

- Before Round 9: \$21.12
- After Round 9: \$26.12
- Gained: \$5 in Round 9

This means in Round 9 (when I was SENDER):

- I sent \$0
- I kept \$5
- I received \$0 back
- Total: \$5

So the receiver I sent to also sent me nothing back, confirming the defection pattern.

****Round 10 (current):****

- I'm RECEIVER
- Sender sent me \$0.02
- I received \$0.06 (0.02×3)
- I can return 0 to 0.06

****Considerations:****

1. This is an interesting change - after multiple rounds of \$0 being sent, someone sent me a tiny amount (\$0.02).
2. This could be:
 - A genuine attempt to restart cooperation (but why so small?)
 - A test to see if I'll reciprocate
 - Communication noise (they might have tried to send more or \$0, but noise affected it)
 - Strategic minimal investment
3. The note says there's communication noise - amounts may differ from what was actually sent/returned.
4. Given the pattern of defection, this tiny amount might be:
 - An attempt to rebuild trust with minimal risk
 - Noise from an intended \$0 send
 - Noise from an intended larger send
5. If I return nothing, I maintain defection
6. If I return something proportional (like \$0.02-0.03), I show willingness to reciprocate even small cooperation
7. If I return a lot (like \$0.06), I show extreme generosity but lose my earnings

****Strategic considerations:****

The sender invested \$0.02 to give me \$0.06. If they were trying to cooperate, a fair return might be:

- Equal split: They keep \$4.98, I keep \$0.03, return \$0.03
- Proportional to standard (1:1 after multiplication): return \$0.02
- Generous (trying to rebuild): return \$0.04-0.06

Given the history of mutual defection and the existence of communication noise, this small gesture could signal a willingness to restart. Returning \$0.02 or \$0.03 demonstrates reciprocity without overcommitting, showing I'm open to cooperation while protecting myself against further defection.

```
Usage: {  
  "input_tokens": 864,  
  "output_tokens": 1084,  
  "reasoning_tokens": 0  
}
```